$$a = \frac{t_x}{s+1/f}\,, \qquad b = \frac{t_z}{s+1/f}\,, \qquad c = \frac{t_y}{s+1/f}\,, \tag{1}$$

$$U = P\exp(\oint \mathcal{A})\,, \tag{2.2}$$

$$U = P\exp(\frac{2\pi}{s+1/f}\ t_+)\,, \tag{A.3}$$

$$S\Psi(r,\theta)S^{-1} = \Psi(r,\theta+2\pi)\,. \tag{4a}$$

$$t_+ = \begin{pmatrix} t_+^{n_1} & & \\ & \ddots & \\ & & t_+^{n_l} \end{pmatrix}, \tag{5}$$

$$\sigma((\lambda^{'};\lambda^{''})) = \sigma(\lambda^{'}) + \sigma(\lambda^{''}). \tag{6.6}$$

$$\alpha_i = f_i - i + 1 \qquad \beta_i = g_i - i + 1. \tag{A.7}$$

$$X_S(m \to m-1) = Y_S. \tag{8a}$$

$$S:\ \ (\lambda,\rho)_G \to (\lambda^{'},\rho^{''})_{G^L}. \tag{9}$$

$$WB:\ \ (\lambda,\emptyset)_B \to (\lambda_{odd}+\lambda_{even},\emptyset) \to (X_S\lambda_{odd},\lambda_{even})_C. \tag{3.10}$$

$$WC:\ \ (\lambda,\emptyset)_C \to (\lambda_{odd}+\lambda_{even},\emptyset) \to (X_S^{-1}\lambda_{even},Y_S\lambda_{odd})_B. \tag{A.11}$$

$$WCC:\ \ (\rho\,;\rho)_C \to (\rho_{\rm even}+\rho_{\rm odd}\,;\rho_{\rm odd}+\rho_{\rm even}) \to (\rho_{\rm even}+X_S^{-1}\rho_{\rm even}\,;\rho_{\rm odd}+Y_S\rho_{\rm odd})_B. \tag{12a}$$

$$CB_{eo}:\ \ (\lambda_{\rm even}\,;\rho_{\rm odd})_C \to (X_S^{-1}\lambda_{\rm even}\,;Y_S\rho_{\rm odd})_B, \tag{13}$$

$$q^9 + 2q^{11} + 4q^{13} + 5q^{15} + 9q^{17} + 12q^{19} + 17q^{21} + 23q^{23} + \cdots \tag{7.14}$$

$$\begin{pmatrix} \alpha_1 & \alpha_2 & \alpha_3 & \cdots \\ \beta_1 & \beta_2 & \cdots & \end{pmatrix}. \tag{A.15}$$

$$\begin{pmatrix} 0 & 0\cdots\cdots 0 & \overbrace{1\cdots\cdots 1}^{m+1} \\ & 0\cdots 0 & \underbrace{1\cdots 1}_{n} \end{pmatrix}, \tag{16a}$$

$$\begin{pmatrix} 0 & 0\cdots\cdots 0 & \overbrace{1\cdots\cdots 1}^{n} \\ & 0\cdots 0 & \underbrace{1\cdots 1}_{m} \end{pmatrix}, \tag{17}$$

$$X_S : \rho_{odd} \to \rho_{even}. \tag{4.18}$$

$$Y_S : \rho_{odd} \to \rho_{even}. \tag{A.19}$$

$$OE : (\lambda_{even}, \rho_{odd})_C \to (X_S^{-1}\lambda_{even}, Y_S\rho_{odd})_B \to (\lambda_{odd}^{'}, \rho_{even}^{'})_B. \tag{20a}$$

$$EO : (\lambda_{odd}^{'}, \rho_{even}^{'})_B \to (X_S\lambda_{odd}^{'}, Y_S^{-1}\rho_{even}^{'})_C \to (\lambda_{even}, \rho_{odd})_C. \tag{21}$$

$$\mathbf{1} : (\lambda, \rho)_D \to (\lambda, \rho)_D, \tag{1.22}$$

$$V_t(x) := V(t, x) : \mathbb{T}^1 \times \mathbb{R} \to \mathcal{S}_n, \tag{A.23}$$

$$x_j(t) := x_j^-(t) + U_1(t)x_j^-(t) \quad \in \ell_1. \tag{24a}$$

$$\langle f, g\rangle_{H^{1/2}, H^{-1/2}} = \langle f, V^* g\rangle_{H^{1/2}} = \langle Vf, g\rangle_{H^{-1/2}}. \tag{25}$$

$$h_t := -\partial_{xx}^2 + V_t(x), \quad \text{acting on} \quad L^2(\mathbb{R}, \mathbb{C}^n), \tag{5.26}$$

$$h_{D,t}^{\sharp} := -\partial_{xx}^2 + V_t \quad \text{acting on} \quad L^2(\mathbb{R}^+, \mathbb{C}^n), \tag{A.27}$$

$$\ell^\circ := \{x \in E, \quad \forall y \in \ell, \quad \omega(x, y) = 0\}. \tag{28a}$$

$$\forall \mathbf{z}_1, \mathbf{z}_2 \in \mathbb{C}^N, \quad \widetilde{\omega}(\mathbf{z}_1, \mathbf{z}_2) = \mathrm{i}\langle \mathbf{z}_1, \mathbf{z}_2\rangle_{\mathbb{C}^N}. \tag{29}$$

$$\langle x, Jy\rangle_{\mathcal{H}_b} = -\langle Jx, y\rangle_{\mathcal{H}_b}, \tag{2.30}$$

$$\mathcal{U} := Q_+^* U Q_- \tag{A.31}$$

$$\mathcal{H}_b := \mathbb{C}^n \times \mathbb{C}^n \tag{32a}$$

$$\omega((\Theta x, \Pi x), (\Theta y, \Pi y)) = \langle \Theta x, \Pi y\rangle_{\mathbb{C}^n} - \langle \Pi x, \Theta y\rangle_{\mathbb{C}^n} = \langle x, (\Theta\Pi - \Pi\Theta)y\rangle = 0, \tag{33}$$

$$\Pi\psi(0) = \Theta\psi'(0). \tag{6.34}$$

$$k \cdot \frac{1 - R(k)}{1 + R(k)} = \frac{1 - \mathcal{U}^+(E)}{1 + \mathcal{U}^+(E)}, \quad k = \sqrt{E}. \tag{A.35}$$

$$\forall x, y \in \ell_1(t^*) \cap \ell_2(t^*), \quad \omega(x, P_t'(y)) = \overline{\omega(y, P_t'(x))}. \tag{36a}$$

$$\delta_{ij}\theta_j'(t^*) = \langle U_2 x_i^-, (\partial_t U_1)x_j^-\rangle_{\mathcal{H}_b}\big|_{t=t^*} = \langle U_1 x_i^-, (\partial_t U_1)x_j^-\rangle_{\mathcal{H}_b}\big|_{t=t^*}. \tag{37}$$

$$h_t := -\partial_{xx}^2 + V_t(x). \tag{3.38}$$

$$\delta_{ij}\lambda_j'(t^*) = \omega\left(u_i, \partial_t\left(P_t^{\sharp}\right)u_j\right)\big|_{t=t^*} - \omega\left(u_i, \left(\partial_t P_t^+\right)u_j\right)\big|_{t=t^*}. \tag{A.39}$$

$$\mathcal{U}(t) = \frac{\cos(\pi t) + \mathrm{i}\sin(\pi t)}{\cos(\pi t) - \mathrm{i}\sin(\pi t)} = \mathrm{e}^{2\mathrm{i}\pi t}. \tag{40a}$$

$$V_t^\chi(x) := V_{L,t}(x)\chi(x) + V_{R,t}(x)(1 - \chi(x)), \tag{41}$$

$$\|(-\Delta)f\|_{\mathcal{H}}^2 = \sum_{\mathbf{k}\in\mathbb{Z}^{d-1}} \left(\|f_{\mathbf{k}}''\|_{L^2(\mathbb{R})}^2 + (4\pi|\mathbf{k}|^2)^2\|f_{\mathbf{k}}\|_{L^2(\mathbb{R})}^2\right) < \infty. \tag{7.42}$$

$$\forall \mathbf{y} \in \Gamma, \quad (\gamma^D \psi)(\mathbf{y}) := \psi(x=0,\mathbf{y}), \quad \left(\gamma^N \psi\right)(\mathbf{y}) = \partial_x \psi(x=0,\mathbf{y}). \tag{A.43}$$

$$\forall f \in H^{3/2}(\Gamma),\ \forall g \in H^{1/2}(\Gamma), \quad \langle f, g\rangle_{L^2(\Gamma)} = \langle f, Ag\rangle_{H^{3/2}} = \langle A^* f, g\rangle_{H^{1/2}}, \tag{44a}$$

$$\mathcal{H} = \mathcal{H}^+ \oplus \mathcal{H}^-, \quad \text{where} \quad \mathcal{H}^\pm := \left\{\psi \in \mathcal{H},\ \psi = 0 \text{ on } \overline{\Omega^\mp}\right\}. \tag{45}$$

$$\Sigma < \inf_{t\in\mathbb{T}^1} \inf \sigma(H_t) \quad \text{and} \quad \Sigma < \inf_{t\in\mathbb{T}^1} \inf \sigma(H_{D,t}). \tag{4.46}$$

$$H_t^\chi - H_t^{\chi_0} = (V_{L,t} - V_{R,t})(\chi - \chi_0), \tag{A.47}$$

$$\widetilde{H}_t^\sharp := -\partial_{xx}^2 + (-\mathrm{i}\partial_y + A(x) + 2\pi t)^2 + V(\mathbf{x}), \quad \text{acting on} \quad L^2\left(\mathbb{R}^+ \times \mathbb{T}^1, \mathbb{C}\right). \tag{48a}$$

$$\widetilde{H}(s) := -\partial_{xx}^2 + \left(-\mathrm{i}\partial_y + bx\right)^2 + sV(x,y). \tag{49}$$

$$0 = (H_D^\sharp - E)\psi_D + (\Sigma - E)\psi_\Sigma, \quad \text{hence} \quad \psi_D = (E-\Sigma)(H_D^\sharp - E)^{-1}\psi_\Sigma. \tag{1.50}$$

$$\widetilde{\ell}(E) = \left\{(f, M(E)f), \quad f \in H^{-1/2}(\Gamma)\right\}, \tag{A.51}$$

$$(H_D^\sharp - E)\psi_D = (E-\Sigma)\psi_\Sigma. \tag{52a}$$

$$(a \le a', b \le b', a \sim b', a' \sim b) \Rightarrow (a \sim b). \tag{53}$$

$$f_p : (i, v) \mapsto (1 - i, v), \tag{5.54}$$

$$\phi(x,x,\dots,x,y) = \phi(x,x,\dots,y,x) = \dots \phi(y,x,\dots,x,x). \tag{A.55}$$

$$\phi(x,x,y) = \phi(x,y,x) = \phi(y,x,x) = x. \tag{56a}$$

$$\phi(f_s(v_1),\dots,f_s(v_k)) = f_s(\phi(v_1,\dots,v_k)). \tag{57}$$

$$\phi(x_1,\dots,x_k) \sim \phi(f_s(y_1),\dots,f_s(y_k)) = f_s(\phi(y_1,\dots,y_k)) \tag{2.58}$$

$$\phi(\sigma(a),\sigma(b),\sigma(c)) = \sigma(\phi(a,b,c)). \tag{A.59}$$

$$\pi_B(v) := (\pi_B(v_1),\dots \pi_B(v_k)) \tag{60a}$$

$$\Phi(v) = (\pi_B(v_1), \phi(\pi_H(v_1),\dots \pi_H(v_k))) \tag{61}$$

$$\phi(v) = \Phi((0,v_1),\dots,(0,v_k)) \tag{6.62}$$

$$\Phi((a_1,x_1),(a_2,x_2),(a_3,x_3)) = (A,X) \tag{A.63}$$

$$(A,X) := \Phi((a_1,x_1),(a_2,x_2),(a_3,x_3)) \sim \Phi((b_1,y_1),(b_2,y_2),(b_3,y_3)) =: (B,Y), \tag{64a}$$

$$A = \min(a_1,a_2,a_3) \sim \min(b_1,b_2,b_3) = B \tag{65}$$

$$\dim_{\mathcal{D}_{\mathbb{F}_p A}} I = \dim_{Q_{\mathrm{ore}}(\mathbb{F}_p A)} Q_{\mathrm{ore}}(\mathbb{F}_p A) \underset{\mathbb{F}_p A}{\otimes} I = 1. \tag{3.66}$$

$$1 \underset{\mathbb{F}_p N}{\otimes} \mathbb{F}_p I_H^N \bigcap \mathcal{D}_{\mathbb{F}_p N} \underset{\mathbb{F}_p N}{\otimes} (u-1) = 1 \underset{\mathbb{F}_p N}{\otimes} \mathbb{F}_p N(u-1). \tag{A.67}$$

$$d(\widetilde{G}_{\widehat{p}}) \leq d\left(\overline{H_1} \coprod \overline{H_2}\right) - 1 = d(\overline{H_1}) + d(\overline{H_2}) - 1. \tag{68a}$$

$$M \cong \frac{\mathbb{F}_p I_{H_1}^G \oplus \mathbb{F}_p I_{H_2}^G}{J}. \tag{69}$$

$$kG \underset{kH}{\otimes} kI_H \longrightarrow kI_H^G \tag{7.70}$$

$$k \otimes_{\mathbb{Z}} K_{\mathrm{ab}} = 0. \tag{A.71}$$

$$J = \{(x, -x) : x \in \mathbb{F}_p I_A^G\} \leq \mathbb{F}_p I_{H_1}^G \oplus \mathbb{F}_p I_{H_2}^G. \tag{72a}$$

$$\dim_{\mathcal{D}_{\mathbb{F}_p G}} M = \dim_{\mathcal{D}_{\mathbb{F}_p H_1}} \mathbb{F}_p I_{H_1} + \dim_{\mathcal{D}_{\mathbb{F}_p H_2}} \mathbb{F}_p I_{H_2} - 1. \tag{73}$$

$$\gamma : \frac{M \oplus R}{(m_0, u)} \longrightarrow M \tag{4.74}$$

$$\gamma^{-1}(m) = (m, 0). \tag{A.75}$$

$$M_0 = \frac{\mathbb{F}_p I_H^N}{\mathbb{F}_p N(u-1)} \tag{76a}$$

$$\left(1 \underset{\mathbb{F}_p N}{\otimes} \mathbb{F}_p N(u-1)\right) t^m = 1 \otimes \mathbb{F}_p\, t^m N(u^{t^{-m}} - 1), \tag{77}$$

$$\mathbb{F}_p I_H^G = \bigoplus_{n \in \mathbb{Z}} \mathbb{F}_p\, t^n I_H^N, \tag{1.78}$$

$$c_{n_2} \otimes t^{n_2+1}(u-1) \in 1 \underset{\mathbb{F}_p N}{\otimes} \mathbb{F}_p t^{n_2+1} I_H^N \bigcap \mathcal{D}_{\mathbb{F}_p N} \underset{\mathbb{F}_p N}{\otimes} t^{n_2+1}(u-1). \tag{A.79}$$

$$c_k \otimes t^k(v-1) - c_{k-1} \otimes t^k(u-1) \in 1 \underset{\mathbb{F}_p N}{\otimes} \mathbb{F}_p t^k I_H^N. \tag{80a}$$

$$w \in 1 \underset{\mathbb{F}_p N}{\otimes} \mathbb{F}_p N[t^{\pm}, \sigma]\,(v - 1 - t(u-1)) = 1 \underset{\mathbb{F}_p N}{\otimes} \mathbb{F}_p G(v - 1 - t(u-1)). \tag{81}$$

$$c \otimes (v - 1 - t(u-1)) \in 1 \underset{\mathbb{F}_p G}{\otimes} \mathbb{F}_p G(v - 1 - t(u-1)), \tag{5.82}$$

$$d(\widetilde{G}_{\widehat{p}}) = d(\widetilde{H_1}_{\widehat{p}}) + d(\widetilde{H_2}_{\widehat{p}}) - 1. \tag{A.83}$$

$$f : \overline{H_1} \coprod \overline{H_2} \longrightarrow \widetilde{G}_{\widehat{p}} \tag{84a}$$

$$\dim_{\mathcal{D}_{\mathbb{F}_p H_1}} \mathbb{F}_p I_{H_1} + \dim_{\mathcal{D}_{\mathbb{F}_p H_2}} \mathbb{F}_p I_{H_2} - 1 = d(H_{1\widehat{p}}) + d(H_{2\widehat{p}}) - 1 = d(\widetilde{G}_{\widehat{p}}). \tag{85}$$

$$\widetilde{G}_{\mathrm{ab}} \cong \frac{\widetilde{H}_{\mathrm{ab}}}{\langle u - v \rangle} \oplus \langle t \rangle. \tag{2.86}$$

$$d(\overline{H}) + 1 = d(\overline{H}) + d(\mathbb{Z}_p) = d\left(\overline{H} \coprod \mathbb{Z}_p\right) > d(\widetilde{G}_{\widehat{p}}). \tag{A.87}$$

$$\dim_{\mathcal{D}_{\mathbb{F}_p G}} M = \dim_{\mathcal{D}_{\mathbb{F}_p H}} \mathbb{F}_p I_H = d(\widetilde{H}_{\widehat{p}}) = d(\widetilde{G}_{\widehat{p}}). \tag{88a}$$

$$\pi_1(\mathcal{H}, \Delta, \Delta_0) \to \pi_1(\mathcal{G}, \Gamma, \Gamma_0) \tag{89}$$

$$0 \to k I_{\widetilde{G}} \to k\widetilde{G} \to k \to 0, \tag{6.90}$$

$$0 \to L \to M \to M/L \to 0 \tag{A.91}$$

$$R \underset{\mathbb{F}_p G}{\otimes} M \to \mathcal{D}_{\mathbb{F}_p G} \underset{\mathbb{F}_p G}{\otimes} M \tag{92a}$$

$$\alpha : M \to R \underset{\mathbb{F}_p G}{\otimes} M \tag{93}$$

$$d(\widetilde{G}_{\widehat{p}}) \geq d(\widetilde{H}_{\widehat{p}}). \tag{3.94}$$

$$d(\widetilde{G}_{\widehat{p}}) = d(\widetilde{H}_{\widehat{p}}). \tag{A.95}$$

$$\mathbb{Q}(\sqrt{d}), \mathbb{Q}(\sqrt{d+1}), \cdots, \mathbb{Q}(\sqrt{d+n}) \tag{96a}$$

$$\mathbb{Q}(\sqrt{d}), \mathbb{Q}(\sqrt{d+b}), \mathbb{Q}(\sqrt{d+2b}), \cdots, \mathbb{Q}(\sqrt{d+nb}) \tag{97}$$

$$F = \{(F_{i-2\epsilon}, L_{i+\epsilon}, F_i) : i \geq 2, \epsilon \in \{\pm 1\}\}, \tag{7.98}$$

$$G = \{(1, 4m^r - 1, m) : m \geq 2, r \geq 1\}, \tag{A.99}$$

$$H = \{(D_1, D_2, m) : \exists\ r, s \in \mathbb{N} \text{ such that } D_1 s^2 + D_2 = \lambda^2 m^r \text{ and } 3D_1 s^2 - D_2 = \pm\lambda^2\}. \tag{100a}$$

$$(2m^{p^{r-1}} - 1) \mid ((m^{p^r - p^{r-1}} - 1) - (2m^{p^{r-1}} - 1)), \text{ i.e., } (2m^{p^{r-1}} - 1) \mid m^{p^{r-1}}(m^{p^r - 2p^{r-1}} - 2). \tag{101}$$

$$(2m^{p^{r-1}} - 1) \mid (m^{p^r - 2p^{r-1}} - 2). \tag{4.102}$$

$$(2m^{p^{r-1}} - 1) \mid (m^{p^r - 2p^{r-1}} - 2) - 2(2m^{p^{r-1}} - 1)). \tag{A.103}$$

$$(2m^{p^{r-1}} - 1) \mid (m^{p^r - np^{r-1}} - 2^{n-1}),\ 1 \leq n \leq p. \tag{104a}$$

$$0 \to \{\text{1d SPT states}\} \xrightarrow{f} \{\text{2d translation invariant SPT states}\} \xrightarrow{g} \{\text{2d SPT states}\} \to 0 \tag{105}$$

$$H^3(G, \mathbb{T}) \oplus H^2(G, \mathbb{T}) \oplus H^2(G, \mathbb{T}) \oplus H^1(G, \mathbb{T}) \tag{1.106}$$

$$\otimes_{\mathrm{stack}} : \mathcal{A}_\Gamma \times \mathcal{A}_\Gamma \to \mathcal{A}_\Gamma^2 \tag{A.107}$$

$$S_0 = \{(\omega, U) \in \mathcal{P}(\mathcal{A}) \times \hom(G, U(\mathbb{C}^d)) | \omega \text{ is SRE, } \omega \circ i_{U(g)} = \omega\ \forall g \in G\} \tag{108a}$$

$$\mathrm{Index}_{2\ \mathrm{trans}}^{\mathcal{A}^2,U\otimes U}(\omega_1 \otimes_{\mathrm{stack}} \omega_2) = \mathrm{Index}_{2\ \mathrm{trans}}^{\mathcal{A},U}(\omega_1) + \mathrm{Index}_{2\ \mathrm{trans}}^{\mathcal{A},U}(\omega_2). \tag{109}$$

$$\mathrm{Index}_{\mathrm{no\ trans}}^{\mathcal{A}^2,U\otimes U}(\omega_1 \otimes_{\mathrm{stack}} \omega_2) = \mathrm{Index}_{\mathrm{no\ trans}}^{\mathcal{A},U}(\omega_1) + \mathrm{Index}_{\mathrm{no\ trans}}^{\mathcal{A},U}(\omega_2). \tag{5.110}$$

$$v^\dagger W_g v W_g^\dagger = K_g^L \otimes K_g^R = K_g \tag{A.111}$$

$$u_R(g,h) \mapsto b_g^R W_g b_h^R W_h W_{gh}^\dagger (b_{gh}^R)^\dagger W_{gh} W_h^\dagger W_g^\dagger u_R(g,h). \tag{112a}$$

$$v^\dagger b_g v K_g b_g^\dagger = v^\dagger b_g v v^\dagger W_g v W_g^\dagger b_g^\dagger = w^\dagger v^\dagger W_g v W_g^\dagger w = w^\dagger K_g w \tag{113}$$

$$\mathrm{Index}_{2\ \mathrm{trans}}^{\mathcal{A},U}(\theta, \tilde{\beta}_g, \eta_g, \alpha_{0,1}, \Theta_1, \omega, \omega_{01}) = \mathrm{Index}_{2\ \mathrm{trans}}^{\mathcal{A},U}(\theta, \tilde{\beta}_g, \eta_g, \alpha_{0,2}, \Theta_2, \omega, \omega_{02}). \tag{2.114}$$

$$\mathrm{Index}_{2\ \mathrm{trans}}(\theta, \tilde{\beta}_{g,1}, \eta_{g,1}, \alpha_0, \Theta, \omega, \omega_0) = \mathrm{Index}_{2\ \mathrm{trans}}(\theta, \tilde{\beta}_{g,2}, \eta_{g,2}, \alpha_0, \Theta, \omega, \omega_0). \tag{A.115}$$

$$k_g^L \otimes k_g^R = \pi_{L_{-1}}(b)^\dagger \tilde{k}_g \pi_{L_{-1}}(b) \tag{116a}$$

$$\phi_{1,R}(\pi_0 \circ \alpha_0 \circ \Theta(A_1) y_1) = \phi_{1,R}(\pi_0 \circ \alpha_0 \circ \Theta(A_2) y_2). \tag{117}$$

$$\phi_1 : \mathrm{IAC}_{L\times R}(\alpha_0, \theta, \Theta) \to \mathcal{U}(\mathcal{H}) : ab \mapsto \phi_{1,L}(a)\phi_{1,R}(b) \tag{6.118}$$

$$\xi(\tau^{-1}(a^\dagger)a) = \tau^{-1}(a^\dagger)a. \tag{A.119}$$

$$\pi_0 \circ \alpha_0 \circ \Theta \circ \gamma_{0;1}^H \circ \gamma_{1;0}^{H_{\mathrm{split}}}(A) = w\phi_1(w^\dagger a_L \otimes a_R w) w^\dagger (w\tilde{a}w^\dagger \tilde{a}^\dagger) \tag{120a}$$

$$= \pi_0 \circ \alpha_0 \circ \Theta \circ \gamma_{0;1}^H \circ \tilde{\eta}_g \circ \beta_g^U \circ \gamma_{1;0}^H. \tag{121}$$

$$v^\dagger b_g^R v h_g K_g^R (b_g^R)^\dagger = \alpha(g) w^\dagger K_g^R w \tag{3.122}$$

$$\mathrm{Index}_{2\ \mathrm{trans}}^{\mathcal{A}^2,U\otimes U}(\omega_1 \otimes_{\mathrm{stack}} \omega_2) = \mathrm{Index}^{\mathcal{A},U}(\omega_1) + \mathrm{Index}_{2\ \mathrm{trans}}^{\mathcal{A},U}(\omega_2). \tag{A.123}$$

$$\tilde{\Phi}(s,I) = \Phi_L(s,I) + \Phi_R(s,I) - \Phi(s,I). \tag{124a}$$

$$(\gamma_{s;t}^H \circ \gamma_{t;s}^{H_L} \otimes \gamma_{t;s}^{H_R})^{-1} = \gamma_{s;t}^{H_L} \otimes \gamma_{s;t}^{H_R} \circ \gamma_{t;s}^H \in \mathrm{GVAut}(\mathcal{A}). \tag{125}$$

$$\tau : \mathcal{A}_\Gamma \to \mathcal{A}_{\tau(\Gamma)} \quad \nu : \mathcal{A}_\Gamma \to \mathcal{A}_{\nu(\Gamma)} \quad \mu : \mathcal{A}_\Gamma \to \mathcal{A}_{\mu(\Gamma)} \tag{7.126}$$

$$\mathrm{Cone}_R(\alpha_0, \theta) \subseteq \mathrm{Cone}_L(\alpha_0, \theta)' \quad \mathrm{Cone}_L(\alpha_0, \theta) \subseteq \mathrm{Cone}_R(\alpha_0, \theta)'. \tag{A.127}$$

$$v^\dagger u_L(g,h) v v^\dagger u_R(g,h) v = v^\dagger (u_L(g,h) \otimes u_R(g,h)) v = v^\dagger W_g W_h W_{gh}^{-1} v. \tag{128a}$$

$$K_g^L \to \delta(g)^{-1} K_g^L \quad K_g^R \to \delta(g) K_g^R \tag{129}$$

$$w^\dagger W_g w = \beta(g) b_g^L b_g^R W_g \quad w^\dagger u_R(g,h) w = \gamma(g,h) b_g^R W_g b_h^R W_h W_{gh}^\dagger (b_{gh}^R)^\dagger W_{gh} W_h^\dagger W_g^\dagger u_R(g,h). \tag{4.130}$$

$$(b_{gh}^R)^\dagger W_{gh} W_h^\dagger W_g^\dagger u_R(g,h) = (b_{gh}^R)^\dagger u_L(g,h)^\dagger = u_L(g,h)^\dagger (b_{gh}^R)^\dagger = W_{gh} W_h^\dagger W_g^\dagger u_R(g,h) (b_{gh}^R)^\dagger \tag{A.131}$$

$$K_g^R \mapsto v^\dagger b_g^R v h_g K_g^R (b_g^R)^\dagger \quad W_g \mapsto w^\dagger W_g w \quad u_R(g,h) \mapsto w^\dagger u_R(g,h) w \tag{132a}$$

$$\begin{aligned} x_1 x_2 &= \pi_0 \circ \alpha_0 \circ \Theta(A_1) y_1 \pi_0 \circ \alpha_0 \circ \Theta(A_2) y_2 = \pi_0 \circ \alpha_0 \circ \Theta(A_1) y_1 \pi_0 \circ \alpha_0 \circ \Theta(A_2) y_1^\dagger y_1 y_2 \\ &= \pi_0 \circ \alpha_0 \circ \Theta(A_1 \xi_1(A_2)) y_1 y_2 \end{aligned} \tag{133}$$

$$w^\dagger \phi_1(K_g^\sigma) w = \phi_1(w^\dagger a_L \otimes a_R K_g^\sigma a_L^\dagger \otimes a_R^\dagger w) = \phi_1(w^\dagger a_\sigma K_g^\sigma a_\sigma^\dagger w). \tag{1.134}$$

$$g(v_n) = e(v_n) \vee b_{q_n} \vee \bigvee \{b(i,n) : i < n, (v_i, v_n) \in E\}. \tag{A.135}$$

$$g(v_i) \wedge \neg g(v_j) = g(v_i) \wedge \neg(g(v_i) \wedge g(v_j)) = \begin{cases} g(v_i) \text{ if } (v_i, v_j) \notin E \\ g(v_i) \wedge \neg b(i,j) \text{ if } (v_i, v_j) \in E \text{ and } i \neq j \end{cases} \tag{136a}$$

$$g(v_n) = b_n \vee \bigvee_{i<n} b_i^n(\varepsilon), \tag{137}$$

$$b_i^n(\varepsilon) = \begin{cases} b_i^n \text{ if } (v_i, v_n) \in E \\ \mathbf{0} \text{ otherwise.} \end{cases} \tag{5.138}$$

$$g(v_{i_k}) \wedge \neg(\bigvee \{g(v_{i_k}) \wedge g(v_{i_j}) : (v_{i_k}, v_{i_j}) \in E\}), \tag{A.139}$$

$$g(v_l) = \bigvee_{\bar{\imath}(|\bar{\imath}|-1)=l, |\bar{\imath}|<n} b_{\bar{\imath}} \vee \bigvee_{\bar{\imath}(|\bar{\imath}|-1)=l, |\bar{\imath}|=n, (v_{i(0)}, \ldots, v_{i(n-1)}) \in E} b_{\bar{\imath}}. \tag{140a}$$

$$\bigwedge_{l=0}^{k-1} g(v_{i(l)}) \neq \mathbf{0} \text{ iff } k < n, \text{ or } k = n \text{ and } (v_{i(0)}, \ldots, v_{i(n-1)}) \in H_n. \tag{141}$$

$$\Phi_{A,C} : \hom_{\mathbf{C}}(F(A), C) \to \hom_{\mathbf{D}}(A, G(C)), \tag{2.142}$$

$$\forall A, B \in \mathrm{Obj}(\mathbf{D}) \; \forall C \in \mathrm{Obj}(\mathbf{C}) \; \forall v \in \hom_{\mathbf{D}}(A,B) \; \forall \psi \in \hom_{\mathbf{C}}(F(B), C) \tag{A.143}$$

$$\mathcal{M} \longrightarrow (B)_{r,d}^A \tag{144a}$$

$$\mathcal{M} \longrightarrow (B)_r^A. \tag{145}$$

$$C \longrightarrow (B)_r^A. \tag{6.146}$$

$$\mathcal{M} \to (B)_r^A \tag{A.147}$$

$$d_e(A, \mathcal{K}) = |\mathrm{Aut}(A)| \cdot d(A, \mathcal{K}). \tag{148a}$$

$$\bar{\imath} \sim_{\mathcal{A}} \bar{\jmath} \Rightarrow h(\bar{\imath}) \sim_{\mathcal{B}} h(\bar{\jmath}). \tag{149}$$

$$\overline{a}_{\bar{\imath}} \equiv_\Delta^{\mathcal{M}} \overline{a}_{\bar{\jmath}}. \tag{3.150}$$

$$\bar{\imath} \sim_{\mathcal{I}} \bar{\jmath} \Rightarrow \overline{a}_{\bar{\imath}} \equiv^{\mathcal{M}} \overline{a}_{\bar{\jmath}}. \tag{A.151}$$

$$(\forall \bar{\jmath})(\mathcal{I} \vDash \eta(\bar{\jmath}) \Rightarrow \mathcal{M} \vDash \varphi(\overline{a}_{\bar{\jmath}})) \tag{152a}$$

$$(\forall \bar{\jmath})(\mathcal{I} \vDash \eta(\bar{\jmath}) \Rightarrow \mathcal{M} \vDash \varphi(\overline{b}_{\bar{\jmath}})) \tag{153}$$

$$\bar{\jmath} \sim_{\mathcal{A}} \bar{\imath} \tag{7.154}$$

$$\overline{b}_{\bar{\jmath}} \equiv_{\Delta}^{\mathcal{M}} \overline{a}_{\bar{\imath}} \tag{A.155}$$

$$(\forall \bar{\imath})(\mathcal{I} \vDash \eta(\bar{\imath}) \Rightarrow \mathcal{M} \vDash \neg\varphi(\overline{a}_{\bar{\imath}})). \tag{156a}$$

$$(\forall \bar{\jmath})(\mathcal{I} \vDash \eta(\bar{\jmath}) \Rightarrow \mathcal{M} \vDash \neg\varphi(\overline{b}_{\bar{\jmath}})), \tag{157}$$

$$(\forall \bar{\imath})(\mathcal{I} \vDash \eta(\bar{\imath}) \Rightarrow \mathcal{M} \vDash \varphi(\overline{a}_{\bar{\imath}})). \tag{4.158}$$

$$X = (\overline{a}_i \mid i \in \mathcal{I}), \tag{A.159}$$

$$Y = (\overline{b}_i \mid i \in \mathcal{I}), \tag{160a}$$

$$\dots a_1 \sim a_3 \sim a_5 \dots \nsim \dots a_2 \sim a_4 \sim a_6 \dots \tag{161}$$

$$\dots b_1 \nsim b_2 \nsim b_3 \dots \tag{1.162}$$

$$K_{X'} + \Delta' = f^*(K_X + \Delta) + E' + \Xi', \tag{A.163}$$

$$K_{X'} + \Delta' = f^*(K_X + \Delta) + E', \tag{164a}$$

$$f^*_{V'}(A|_V) = A'|_{V'} + E_0 \tag{165}$$

$$\operatorname{mult}_{P'}(E'|_{V'} - t_0 E_0) = a\big(P', V, (\Delta + t_0 A)|_V\big) \geq 0. \tag{5.166}$$

$$K_{X'} + \Delta' - T' + t_0 A' \sim_{\mathbb{Q}, Z'} K_{X'} + \Delta' + t_0 A'. \tag{A.167}$$

$$(X', \Delta' + t_0 A') \dashrightarrow (X'', \Delta'' + t_0 A'') \tag{168a}$$

$$\pi_Y^* \eta = \sum_i \mu_i G_i, \tag{169}$$

$$h^* D = \sum_j a_j D_j, \tag{2.170}$$

$$(X, \Delta + tB) \dashrightarrow (X', \Delta' + tB') \tag{A.171}$$

$$A' \sim_{\mathbb{Q}, Z} B' \sim_{\mathbb{Q}, Z} \frac{1}{t}(K_{X'} + \Delta' + tB') \tag{172a}$$

$$n := mn' \prod_{i=0}^{d} n''(i,\, n'). \tag{173}$$

$$S = k_B \ln \Omega_m, \tag{6.174}$$

$$\mathcal{L} = 2P\frac{d\mathcal{H}}{dP} - \mathcal{H} \tag{A.175}$$

$$\mathcal{H}(P) \approx -\Lambda - P + \mathcal{O}(P^2). \tag{176a}$$

$$\mathcal{L} = \frac{\Lambda(1 - \frac{10P}{\Lambda})}{(1 - \frac{6P}{\Lambda})^{3/2}}. \tag{177}$$

$$F = dA = -A(r)'dt \wedge dr \;\; and \;\; {}^*F = E(r)r^2 sin\theta(d\theta \wedge d\varphi), \tag{3.178}$$

$$ds^2 = -f(r)dt^2 + \frac{dr^2}{f(r)} + r^2 d\theta^2 + r^2 \sin^2\theta d\varphi^2. \tag{A.179}$$

$$E(r) = \frac{\sqrt{6}}{q}\left\{\frac{4(1-2A)^2 + 48(1-2A)Br + 27B^2r^2 + (-1+2A)kr^4}{(24 - 48A + 24Br + kr^4)^{3/2}}\right\}. \tag{180a}$$

$$f(r) = 1 \pm \sqrt{\frac{4M\Lambda r}{3} + \frac{\Lambda^2 r^4}{9} - \frac{2q^2\Lambda}{3}}. \tag{181}$$

$$f(r) = 1 - \sqrt{\frac{r^4}{4q^4} - \frac{2Mr}{q^2} + 1}. \tag{7.182}$$

$$S = \pi r_H^2. \tag{A.183}$$

$$M(S, q, \Lambda) = 2S\left(\frac{\partial M}{\partial S}\right) + q\left(\frac{\partial M}{\partial q}\right) - 2\Lambda\left(\frac{\partial M}{\partial \Lambda}\right). \tag{184a}$$

$$\Phi_q = \frac{\partial M}{\partial q} = \sqrt{\frac{\pi}{S}}q. \tag{185}$$

$$T_H = \frac{3 - 2q^2\Lambda - \Lambda^2 r_H^4}{8\pi\Lambda r_H^3}. \tag{4.186}$$

$$C_q = \frac{2\pi}{3}\left(\frac{3r_H^6 - 8r_H^2 q^4}{8q^4 + r_H^4}\right). \tag{A.187}$$

$$E = \int T_H dS. \tag{188a}$$

$$\frac{dV}{dr_H} = 4\pi r_H^2 \tag{189}$$

$$H = E + PV. \tag{1.190}$$

$$G = \frac{2q^2}{r_H}. \tag{A.191}$$

$$S = S_0 - \frac{\alpha}{2} lnCT_H^2 + (sub \;\; leading \;\; terms), \tag{192a}$$

$$P = -\frac{dF}{dV}, \tag{193}$$

$$\Psi\left(t\right) = \exp(-it\sqrt{A_E})\Psi\left(0\right). \tag{5.194}$$

$$f(r) \approx 1 + \sqrt{\eta - \beta r} \tag{A.195}$$

$$0 \leq \frac{r^{1+p}}{1+\sqrt{\eta-\beta r}} \leq \frac{r^{1+p}}{\sqrt{\eta-\beta r}}. \tag{196a}$$

$$\int_{const.}^{0} \frac{r^{1-p}}{1+\eta-\beta r} dr = \frac{r^{2-p}{}_2F_1\left(1, 2-p; 3-p; \frac{\beta r}{1+\eta}\right)}{(1+\eta)(2-p)}|_{const}^{0}, \tag{197}$$

$$R(r) = c_3 + \frac{c_4}{r^3}, \tag{2.198}$$

$$c_4^2 \int_{const.}^{\infty} \frac{1}{r^4(1+\delta r^2)} dr = c_4^2 \left[\frac{\delta}{r} - \frac{1}{3r^3} + \delta^{3/2} tan^{-1}(\sqrt{\delta} r)\right] |_{const}^{\infty} < \infty. \tag{A.199}$$

$$\psi'' - \frac{\beta}{2\bar{\eta}}\psi' - \frac{\bar{\lambda}^2}{r^2}\psi = 0, \tag{200a}$$

$$\|R\|^2 \sim \int_{const.}^{0} \frac{r^2|R|^2}{1+\sqrt{\eta-\beta r}} dr = \bar{c_6}^2 \int_{const.}^{0} \frac{r^{3-2\sigma}}{1+\sqrt{\eta-\beta r}} dr. \tag{201}$$

$$\psi^{''} + \frac{2}{r}\psi^{'} = 0, \tag{6.202}$$

$$2c_7c_8 \int_{const.}^{\infty} \frac{r}{1+\delta r^2} dr = c_7c_8 \left[\frac{1}{\delta} ln(\delta r^2 + 1)\right] |_{const}^{\infty} \to \infty, \tag{A.203}$$

$$H \,=\, \sum_{a=0}^{n-1} \frac{q_a}{r_a}\,, \qquad K^i \,=\, \sum_{a=0}^{n-1} \frac{k_a^i}{r_a}\,, \qquad L^i \,=\, l_0^i + \sum_{a=0}^{n-1} \frac{l_a^i}{r_a}\,, \qquad M \,=\, m_0 + \sum_{a=0}^{n-1} \frac{m_a}{r_a}\,, \tag{204a}$$

$$P \,=\, 1 + \sum_{a=0}^{n-1} \frac{\lambda_a}{r_a}\,, \qquad Q \,=\, \sum_{a=0}^{n-1} \frac{\sigma_a\lambda_a}{r_a}\,. \tag{205}$$

$$ds_4^2 \,=\, H^{-1}(d\psi + A)^2 + H ds_3^2\,, \tag{3.206}$$

$$k_a^0 \,=\, \frac{g_s\alpha'}{2R_y} n_a^0\,, \qquad k_a^1 \,=\, \frac{g_s\alpha'^3}{2V_4R_y} n_a^1\,, \qquad k_a^2 \,=\, \frac{R_y}{2} n_a^2\,, \tag{A.207}$$

$$\sum_{b\neq a} \frac{q_aq_b}{r_{ab}} \Pi_{ab}^0 \Big(\Pi_{ab}^1\Pi_{ab}^2 - \frac{1}{2g^2}\mathbb{T}_{ab}\Big) = \sum_{b,i} q_aq_b l_0^i \Pi_{ab}^i\,, \tag{208a}$$

$$\Pi_{ab}^i = \frac{k_b^i}{q_b} - \frac{k_a^i}{q_a}, \qquad \mathbb{T}_{ab} = \frac{1}{q_a^2} + \frac{1}{q_b^2}\,. \tag{209}$$

$$\sum_{b\neq a}\frac{q_a q_b}{r_{ab}}\Pi^0_{ab}\left(\Pi^1_{ab}\Pi^2_{ab}-\frac{1}{2g^2}\mathbb{T}_{ab}\right)\,=\,0\,. \tag{7.210}$$

$$\mathcal{M}^2_{\alpha\beta}\Pi^2_{1(\alpha+1)}\,\equiv\,\Big(\bar{\mathcal{M}}^2_{\alpha\beta}+\lambda\dot{\mathcal{M}}^2_{\alpha\beta}\Big)\Pi^2_{1(\alpha+1)}\,=\,\bar{B}^2_\beta+\lambda\dot{B}^2_\beta\,. \tag{A.211}$$

$$r^i_a\,\to\,\bar{r}^i_a\,\equiv\,\lambda\, r^i_a\,,\qquad \lambda\ll 1\,, \tag{212a}$$

$$\sum_{b\neq n-1}\frac{q_{n-1}q_b}{r_{(n-1)b}}\Pi^0_{(n-1)b}\left(\Pi^1_{(n-1)b}\Pi^2_{(n-1)b}-\frac{1}{2g^2}\mathbb{T}_{(n-1)b}\right)\,=\,0\,. \tag{213}$$

$$q_{n-1}q_b\Pi^0_{(n-1)b}\left(\Pi^1_{(n-1)b}\tilde{\Pi}^2_{(n-1)b}-\frac{1}{2g^2}\mathbb{T}_{(n-1)b}\right). \tag{4.214}$$

$$r^i_a\,\sim\,U\big(-1,1\big)\,. \tag{A.215}$$

$$\{r^{(p)}_A,\sigma^{(p)}_A\}\,,\qquad A=(a,i)\,. \tag{216a}$$

$$\sum_{a\neq b}\frac{q_a q_b}{r_{ab}}\Pi^0_{ab}\Big(\Pi^1_{ab}\Pi^2_{ab}-\frac{1}{2g^2}\mathbb{T}_{ab}\Big)-\sum_{b,i}q_a q_b l^i_0\Pi^i_{ab}\,=\,\epsilon_a\,, \tag{217}$$

$$f(r_A)\,=\,\frac{1}{\sum_a|\epsilon_a|}\,. \tag{1.218}$$

$$f_{\rm inv}(r_A)\,=\,\sum_a|\epsilon_a|\,. \tag{A.219}$$

$$\tilde{f}(r_A)\,=\,\frac{1}{\max_a|\epsilon_a|}\,, \tag{220a}$$

$$\tilde{f}_{\rm inv}(r_A)\,=\,\max_a|\epsilon_a|\,. \tag{221}$$

$$r^{(p)}_A\,=\,\bar{r}_A+\delta r^{(p)}_A\qquad \text{for } p=1,\cdots,\texttt{pop_size}\,. \tag{5.222}$$

$$r'^{(p)}_A=2\bar{r}_A-r^{(p)}_A\,, \tag{A.223}$$

$$P^{(p)}=\frac{f(r^{(p)}_A)}{\sum_p f(r^{(p)}_A)}\,, \tag{224a}$$

$$f'(r^{(p)}_A)=\max\Big(f\big(r^{(p)}_A\big)-\big(\bar{f}-c\sigma_f\big),0\Big)\,, \tag{225}$$

$$\sigma_A\,\to\,\sigma'_A\,=\,\sigma_A\exp\Big(\tau'\,\delta\sigma+\tau\,\delta\sigma_A\Big)\,, \tag{2.226}$$

$$r_A\,\to\,r'_A\,=\,r_A+\sigma'_A\,\delta r_A\,. \tag{A.227}$$

$$\hat{r}_A\,=\,\frac{1}{\texttt{pop_size}}\sum_q r^{(q)}_A\,, \tag{228a}$$

$$
\begin{aligned}
&Q_1 \simeq 2938\,, \qquad Q_5 \simeq 2016\,, \qquad Q_P \simeq 2.998 \times 10^4\,, \\
&J_L \simeq 4.090 \times 10^5\,, \qquad J_R \simeq 0.001545\,,
\end{aligned}
\tag{229}
$$

$$
\begin{aligned}
&Q_1 \approx 736.3\,, \qquad Q_5 \approx 410.0\,, \qquad Q_P \approx 8.887 \times 10^4\,, \\
&J_L \approx 1.625 \times 10^5\,, \qquad J_R \approx 0.007047\,.
\end{aligned}
\tag{6.230}
$$

$$
S = \alpha L - \beta \log L - \gamma + o(L^{-1}).
\tag{A.231}
$$

$$
h_v = -\sum_{a=1}^{3} A_v^{(a)} + \hat{\Delta}_v,
\tag{232a}
$$

$$
A_v^{(a)} = \prod_{j \in +_v} Z_j^{(a)}, \quad \text{for } a = 1, 2, 3,
\tag{233}
$$

$$
h_f = -\sum_{a=1}^{3} B_f^{(a)} - \hat{N}_f,
\tag{3.234}
$$

$$
H = \sum_v h_v + \sum_f h_f
\tag{A.235}
$$

$$
C_{<f,f'>}^{(a)} = \prod_{j \in (f \cup f') \setminus (f \cap f')} X_j^{(a)}, \quad \text{for } a = 1, 2, 3.
\tag{236a}
$$

$$
H_{\text{int}} = \sum_v h_v + \sum_f h_f + \sum_{<f,f'>} h_{<f,f'>}
\tag{237}
$$

$$
h_{<f,f'>} = -\sum_{a=1}^{3} C_{<f,f'>}^{(a)} - \hat{N}_{<f,f'>},
\tag{7.238}
$$

$$
[A_v^{(a)}, B_f^{(a)}] = 0, \qquad \forall v, f,
\tag{A.239}
$$

$$
[B_f^{(a)}, B_f^{(b)}] \neq 0,
\tag{240a}
$$

$$
\begin{aligned}
[A_v^{(a)}, B_f^{(b)}] &= 0, \\
[A_v^{(a)}, C_{<f,f'>}^{(b)}] &= 0,
\end{aligned}
\tag{241}
$$

$$
\begin{aligned}
H_v &= \sum_v h_v, \\
H_f &= \sum_f h_f + \sum_{<f,f'>} h_{<f,f'>},
\end{aligned}
\tag{4.242}
$$

$$
[H_v, H_f] = 0.
\tag{A.243}
$$

$$
(1 + s_{\text{r}} + s_{\text{b}})^{2l} = \sum_{i+j=0}^{2l} \binom{2l}{i \quad j \quad 2l-i-j} s_{\text{r}}^i s_{\text{b}}^j,
\tag{244a}
$$

$$N_{\mathrm{i}} = \sum_{i,j=0 \mod 2} \binom{2l}{i \quad j \quad 2l-i-j} = \frac{3^{2l}+3}{4}. \tag{245}$$

$$S_{\mathrm{i}} = -\sum p_\sigma \log p_\sigma = 2l \log 3 - \log 4 + \epsilon, \tag{1.246}$$

$$h'_v = -\sum_{a=1}^{2} {A'}_v^{(a)} + \hat{\Delta}'_v, \tag{A.247}$$

$$h'_f = -\sum_{a=1}^{2} {B'}_f^{(a)} + \hat{U}_f^{(0)}, \tag{248a}$$

$$T = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{pmatrix}, \tag{249}$$

$$S_{\mathrm{n}} = 2l \log(1+\sqrt{3}) + \epsilon. \tag{5.250}$$

$$N_{\mathrm{FPL}} \approx \frac{2^{2l}}{\sqrt{\pi l}}. \tag{A.251}$$

$$S_{\mathrm{FPL}} = 2l \log 2 - \frac{1}{2}\log(2l) - \frac{1}{2}\log\frac{\pi}{2}, \tag{252a}$$

$$N_{\mathrm{BLC}} = \sum_{n=0}^{l} \binom{2l}{n \quad n \quad 2l-2n} s^n. \tag{253}$$

$$\frac{\sigma}{2} = \frac{2s-\sqrt{s}}{4s-1} \equiv \sqrt{\sigma}(1-\sigma). \tag{2.254}$$

$$W_{x,y}^{(a)} = \prod_{j \in \mathcal{C}_{x,y}} X_j^{(a)}, \quad \text{and} \quad \tilde{W}_{x,y}^{(a)} = \prod_{j \in \tilde{\mathcal{C}}_{x,y}} Z_j^{(a)}, \tag{A.255}$$

$$\{W_x^{(a)}, \tilde{W}_y^{(a)}\} = 0, \quad \text{and } \{W_y^{(a)}, \tilde{W}_x^{(a)}\} = 0, \tag{256a}$$

$$\sigma_{DC} = \sigma_0 + \tilde{\sigma}, \tag{257}$$

$$\tan\Theta_H \sim \frac{B}{n}\tilde{\sigma}. \tag{6.258}$$

$$\sigma_0 \propto T^{-1}, \qquad \tilde{\sigma} \propto T^{-2}, \qquad n = \text{const}. \tag{A.259}$$

$$
\begin{aligned}
U(z) &= \frac{1+(1+3Q)z+z^2(1+3Q(1+Q)-\frac{1}{2}k^2)}{(1+Qz)^{3/2}}\,,\\
V(z) &= (1+Qz)^{3/2}\,,\\
a(z) &= \frac{\sqrt{3Q(1+Q)\left(1-\frac{k^2}{2(1+Q)^2}\right)}}{1+Qz}\,,\quad \varphi(z)=Q\,.
\end{aligned}
\tag{260a}
$$

$$
\rho_{xx}=\frac{\sigma_{xx}}{\sigma_{xx}^2+\sigma_{xy}^2},
\tag{261}
$$

$$
\cot\Theta_H=\frac{\sigma_{xx}}{\sigma_{xy}}.
\tag{3.262}
$$

$$
\rho_{xx}\propto T^{\gamma}\,,\quad \cot\Theta_H\propto T^{\gamma}\,,
\tag{A.263}
$$

$$
\tilde{\sigma}=\frac{n^2}{\chi_{\pi\pi}}\,\frac{\Omega}{\Omega\,\Gamma+\omega_0^2}\,,
\tag{264a}
$$

$$
\tilde{\sigma}=\frac{n^2}{\chi_{\pi\pi}\,\Gamma}\,,
\tag{265}
$$

$$
\rho_{xx}\propto T\,,\qquad \cot\Theta_H\propto T^{\beta}\,,
\tag{7.266}
$$

$$
\sigma_0\sim T^{-x_1}\,,\quad \tilde{\sigma}\sim T^{-x_2}\,,\quad n\sim T^{x_3}\,,\quad B\sim T^0\,.
\tag{A.267}
$$

$$
\beta=x_2+x_3.
\tag{268a}
$$

$$
\rho_{xx}\propto\frac{T^2}{W+T}\,,
\tag{269}
$$

$$
H_{\mathrm{XY}}=-\sum_{j=0}^{L-1}\left(\sigma_j^-\sigma_{j+1}^+ +\sigma_j^+\sigma_{j+1}^-\right)
\tag{4.270}
$$

$$
\mathcal{G}_N(t)=\det_{1\le a,b\le N}\left[i^{b-a}J_{a-b}(-2t)\right],
\tag{A.271}
$$

$$
\mathcal{R}_N(\ell,\tau)=1-\frac{f(\tau)}{f_{N,L}(\tau)},
\tag{272a}
$$

$$
i\frac{\mathrm{d}\mathcal{G}_{N,L}(t;\kappa)}{\mathrm{d}t}=\sum_{a=1}^{N}\left[\mathcal{G}_{N,L}(t;\kappa+\square_a)+\mathcal{G}_{N,L}(t;\kappa-\square_a)\right].
\tag{273}
$$

$$
\mathrm{supp}\rho=[-\pi,\pi].
\tag{1.274}
$$

$$
\rho(x)=\begin{cases}\frac{1+2\gamma x}{\pi\sqrt{1-x^2}}\mathbb{1}_{-1<x<1} & 0\le\gamma\le\frac{1}{2}\\ \frac{2\gamma}{\pi}\sqrt{\frac{x-A}{1-x}}\mathbb{1}_{A\le x\le 1} & \gamma>\frac{1}{2},\end{cases}
\tag{A.275}
$$

$$
\rho(x)=i\frac{2\tau}{\pi}\sqrt{\frac{(A-x)(x-B)}{1-x^2}}
\tag{276a}
$$

$$\frac{\mathfrak{D}^{(p)}_{N+1,\infty}(t)}{\mathcal{G}_{N,\infty}(t)} = \left\langle \chi_{\mathsf{A}_p} \right\rangle_{N,t} \tag{277}$$

$$\mathcal{G}^2_{N+1} \frac{\partial^2}{\partial t^2} \ln \mathcal{G}_{N+1} \approx -4 \mathcal{G}_N \mathcal{G}_{N+2}, \tag{5.278}$$

$$\lim_{\tau \to \tau_{\rm cr}} \partial^2_\tau f(\tau) = 2, \qquad \lim_{\tau \to \tau_{\rm cr}} \partial^3_\tau f(\tau) \neq 0, \tag{A.279}$$

$$k_1 < k_2 < \cdots < k_N. \tag{280a}$$

$$|\psi_0\rangle := |N, \emptyset\rangle. \tag{281}$$

$$\mathcal{L}_{N,\infty}(t) = \mathcal{G}_{N,\infty}(t) \mathcal{G}_{N,\infty}(-t) \tag{2.282}$$

$$\mathsf{a} = \frac{a}{N}, \qquad 0 < \mathsf{a} \leq 1. \tag{A.283}$$

$$\mathcal{G}_{N,\infty}(-i\beta) = \langle \psi_0 | e^{-\beta H_{\rm XY}} | \psi_0 \rangle. \tag{284a}$$

$$\cot\left(\frac{\theta - \varphi}{2}\right) = 2 \sum_{k=1}^{\infty} \left[\cos(k\theta) \sin(k\varphi) - \sin(k\theta) \cos(k\varphi) \right], \tag{285}$$

$$\frac{\mathrm{d}\mathcal{F}^{(\rm PBC)}}{\mathrm{d}\gamma} = 2 \int \mathrm{d}\theta \rho(\theta) \cos(\theta) = \begin{cases} 2\gamma & 0 \leq \gamma \leq \frac{1}{2} \\ 2 - \frac{1}{2\gamma} & \gamma \geq \frac{1}{2}. \end{cases} \tag{6.286}$$

$$\mathcal{F}^{(\rm PBC)}|_{N \mapsto 2N+1} = 2 \mathcal{F}^{(\rm ABC)}|_N. \tag{A.287}$$

$$0 \leq \frac{1}{\pi} \left[\mathrm{ArcSin}(z) - \mathrm{ArcSin}(-1) \right] \leq 1, \tag{288a}$$

$$\lim_{\tau \to \tau_{\rm cr}} A = \lim_{\tau \to \tau_{\rm cr}} B = \frac{i}{2\tau_{\rm cr}} \tag{289}$$

$$\lim_{\tau \to 0^+} \Gamma = \mathbb{S}^1. \tag{3.290}$$

$$\mathfrak{D}(w) = \sum_{p=0}^{N} w^p \left\langle \chi_{\mathsf{A}_p} \right\rangle. \tag{A.291}$$

$$|\rho(z)| \leq \left| \rho^{\mathscr{D}}_{\rm unif.} \right| \tag{292a}$$

$$\gamma = \gamma_\star \equiv \frac{\ell - 1}{2}, \tag{293}$$

$$\frac{\sqrt{\ell_\star^2 - 1}}{2} = \tau_{\rm cr} \quad \Longrightarrow \quad \ell_\star \approx 1.2 \tag{7.294}$$

$$\tilde{\nu}(x) := \sqrt{1 - x^2} \frac{\nu'(x)}{iL\nu(x)} = -\frac{\nu(x) - 1}{\nu(x)} = \frac{z^L}{1 - z^L}. \tag{A.295}$$

$$p_j(x) \approx \frac{q(x)^{j+\frac{1}{2}}}{(x^2-1)^{\frac{1}{4}}} e^{\frac{it}{q(x)}}, \qquad j \to \infty, \tag{296a}$$

$$\lim_{N\to\infty} \ln \mathcal{G}_N(t) = -2t^2 \tag{297}$$

$$\lim_{n\to\infty} \frac{1}{(2n+1)^2} \ln\left[-\frac{1}{4}\mathcal{G}_{2n+1}^2 \frac{\partial^2}{\partial t^2} \ln \mathcal{G}_{2n+1}\right] = \lim_{n\to\infty} \frac{1}{(2n+1)^2} \ln\left[\mathcal{G}_{2n}\mathcal{G}_{2n+2}\right]. \tag{4.298}$$

$$\begin{aligned} \text{imaginary time}: \qquad & \gamma := \frac{\beta}{N} \text{ fixed}; \\ \text{real time}: \qquad & \tau := \frac{t}{N} \text{ fixed}. \end{aligned} \tag{A.299}$$

$$Z_A = \sum_{\{s_1\},\{s_2\}} W_{s_1} W_{s_2} \det g_A^{s_1,s_2}, \tag{300a}$$

$$\mathcal{Z}(\lambda) = \sum_{C\subseteq A} \lambda^{N_C} (1-\lambda)^{N_A - N_C} Z_C, \tag{301}$$

$$\frac{\mathcal{Z}(\lambda_j)}{\mathcal{Z}(\lambda_i)} = \left\langle \left(\frac{\lambda_j}{\lambda_i}\right)^{N_C} \left(\frac{1-\lambda_j}{1-\lambda_i}\right)^{N_A - N_C} \right\rangle_{\lambda_i}, \tag{1.302}$$

$$S_2 = \mathcal{A}L - 3a_2^{\mathrm{SM}}(\pi/3)\ln(L) + \text{const.} \tag{A.303}$$

$$S_2 = \mathcal{A}L + \frac{N_g}{2}\ln(L) + \text{const.} \tag{304a}$$

$$G^{\tilde{s}}(\theta) = G^s(\theta) + \gamma_i G^s(\theta,\tau)\boldsymbol{v}_i \boldsymbol{v}_i^T G^s(\tau,\theta). \tag{305}$$

$$R = \frac{W_{s_1'}}{W_{s_1}} \frac{\det(g_C^{s_1',s_2})}{\det(g_C^{s_1,s_2})}. \tag{5.306}$$

$$R_\uparrow = 1 + (e^{2i\alpha\tilde{\sigma}_i} - 1)(1 - G_{i,i}^{s_1}(\tau) - \Gamma), \tag{A.307}$$

$$(g_C^{\tilde{s}_1,s_2})^{-1} = (g_C^{s_1,s_2})^{-1} + \rho_i \boldsymbol{a}_i \boldsymbol{b}_i^T, \tag{308a}$$

$$P_{\text{join}} = \min\left\{1, \frac{\lambda}{1-\lambda}\frac{\det(g_{C+i}^{s_1,s_2})}{\det(g_C^{s_1,s_2})}\right\} \tag{309}$$

$$P_{\text{split}} = \min\left\{1, \frac{1-\lambda}{\lambda}\frac{\det(g_{C-i}^{s_1,s_2})}{\det(g_C^{s_1,s_2})}\right\} \tag{2.310}$$

$$d(gh_1, gh_2) = d(h_1,h_2), \quad g,h_1,h_2 \in G. \tag{A.311}$$

$$U : G \to \mathrm{U}(\mathcal{H}) \tag{312a}$$

$$U(\mathrm{e}^X) = \mathrm{e}^{-\mathrm{i}A(X)}, \quad \text{for all } X \in \mathfrak{g}. \tag{313}$$

$$\Delta = \sum_j [A(X_j)]^2, \tag{6.314}$$

$$c\Delta' \leq \Delta \leq C\Delta', \tag{A.315}$$

$$A(X) = \frac{\mathrm{i}}{4}\, c \cdot Xc \tag{316a}$$

$$\|\mathcal{U}_g - \mathcal{U}_h\|_\diamond \leq \sqrt{m(2m-1)/2}\; d(g,h). \tag{317}$$

$$A(X) = \frac{1}{2} R \cdot \Omega X R \tag{3.318}$$

$$U : \mathrm{Mp}(2m, \mathbb{R}) \to \mathrm{U}(\mathcal{H}). \tag{A.319}$$

$$\mathrm{e}^{\alpha h} = \begin{pmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \cos(\alpha) \end{pmatrix}. \tag{320a}$$

$$A(h) = -\frac{1}{2} R \cdot \Omega^2 R = \frac{1}{2}(Q^2 + P^2) \tag{321}$$

$$\|\mathcal{U}_g - \mathcal{U}_h\|_\diamond^{H^2,E} \leq 2\sqrt{E + \frac{3m}{8}}\; d(g,h). \tag{7.322}$$

$$[X_0, X_1] = X_2, \quad [X_0, X_2] = -X_1 \quad \text{and} \quad [X_1, X_2] = -X_0. \tag{A.323}$$

$$D = N_1 - N_2. \tag{324a}$$

$$[K_0, K_1] = \mathrm{i}K_2, \quad [K_0, K_2] = -\mathrm{i}K_1 \quad \text{and} \quad [K_1, K_2] = -\mathrm{i}K_0. \tag{325}$$

$$(p, q) \mapsto \eta(p, q) = \vec{p} \cdot \vec{q} - p_0 q_0, \tag{4.326}$$

$$\Lambda^\top \eta \Lambda = \eta. \tag{A.327}$$

$$U_\Lambda \psi(p) = \psi(\Lambda^{-1} p). \tag{328a}$$

$$(M_{mn})_{ij} = \eta_{ni}\delta_{mj} - \eta_{mi}\delta_{nj}. \tag{329}$$

$$\sum_j A(Y_j)^2 = \sum_{jkl} J_{jk} J_{jl}\, A(X_k) A(X_l) = \sum_{kl} \delta_{kl}\, A(X_k) A(X_l) = \Delta. \tag{1.330}$$

$$\omega \equiv \omega(\rho, \psi, p) = \begin{pmatrix} (1-p)\rho & |\psi\rangle \\ \langle\psi| & p \end{pmatrix} \tag{A.331}$$

$$\omega^{(k)} = \begin{pmatrix} (1-p)\langle k|\rho|k\rangle & \langle k|\psi\rangle \\ \langle\psi|k\rangle & p \end{pmatrix} \geq 0. \tag{332a}$$

$$\Delta_D = \frac{1}{4}\left(\sum_{k=1}^m Q_k^4 + \sum_{k=1}^m P_k^4 \right). \tag{333}$$

$$A(B_{jk}) = \frac{\mathrm{i}}{2\sqrt{2}} c_k c_j, \quad j = 1, \ldots, 2m, \quad k < j. \tag{5.334}$$

$$c_{2j-1} = b_j^* + b_j \quad \text{and} \quad c_{2j} = -\mathrm{i}(b_j^* - b_j). \tag{A.335}$$

$$b_j = \frac{1}{2}(c_{2j-1} - \mathrm{i}c_{2j}) \quad \text{and} \quad b_j^* = \frac{1}{2}(c_{2j-1} + \mathrm{i}c_{2j}). \tag{336a}$$

$$\begin{aligned} \|\mathcal{U}_g - \mathcal{U}_h\|_\diamond &\leq \sqrt{m(2m-1)/2}\, \|\log(g^{-1}h)\|_2 \\ &\leq \frac{\pi}{2\sqrt{2}} \sqrt{m(2m-1)}\, \|g - h\|_2, \end{aligned} \tag{337}$$

$$\begin{aligned} M_{k,\ell} &= (E_{k\ell} + E_{\ell k})/\sqrt{2}, \quad (\ell = 2, \ldots, 2m; k < \ell), \\ D_k &= E_{kk}, \quad (k = 1, \ldots, 2m). \end{aligned} \tag{2.338}$$

$$B_{jk} = (E_{jk} - E_{kj})/\sqrt{2}, \quad j = 1, \ldots, 2m, \quad k < j, \tag{A.339}$$

$$A(B_{jk}) = \frac{\mathrm{i}}{4\sqrt{2}}[c_j, c_k], \quad j = 1, \ldots, 2m, \quad k < j. \tag{340a}$$

$$\mathrm{N}_g = \mathrm{U}_g e^{\Lambda} e^{\Xi}, \tag{341}$$

$$\Xi := i\epsilon \sum_{k=1}^{N^2-1} \int_0^1 \mathrm{dW_{k,s}} \mathrm{L}_{k,s}. \tag{6.342}$$

$$\xi_{kij}^+ = \int_0^1 \mathrm{dW}_{k,s} \mathrm{L}_{kij,s}^+, \quad \xi_{kij}^- = \int_0^1 \mathrm{dW}_{k,s} \mathrm{L}_{kij,s}^-, \tag{A.343}$$

$$\mathbb{E}\big[\xi_{kij}^\pm \xi_{k'i'j'}^\pm\big] = \delta_{k,k'} \int_0^1 \mathrm{d}s \mathrm{L}_{kij,s}^\pm \mathrm{L}_{ki'j',s}^\pm. \tag{344a}$$

$$\mathrm{H}^{(1,2)}(\theta, \phi) = \frac{\hbar\theta}{2} \mathrm{Z}^{(1)} \otimes \mathrm{R}_{xy}^{(2)} \tag{345}$$

$$\sigma_s^{\pm(1)} = e^{\pm is\theta \mathrm{R}_{xy}^{(2)}(\phi)} \sigma^{\pm(1)}, \tag{3.346}$$

$$\mathcal{R}(\theta, \phi) = \cos(\theta/2)\mathrm{Z}^{(2)} + \sin(\theta/2)\mathrm{Z}^{(1)}\mathrm{R}_{xy}^{(2)}(\phi). \tag{A.347}$$

$$\xi_{k,+}^{(i)} = \int_0^1 \mathrm{dW}_{k,s}^{(i)} \cos(s\theta), \quad \xi_{k,-}^{(i)} = \int_0^1 \mathrm{dW}_{k,s}^{(i)} \sin(s\theta), \tag{348a}$$

$$\xi_{k,w}^{(i)} = \int_0^1 \mathrm{dW}_{k,s}^{(i)}, \tag{349}$$

$$f_w = \frac{1}{2}\big[\epsilon_1 e^{-i\phi} \xi_{1,w}^{(2)} + \epsilon_2 e^{i\phi} \xi_{2,w}^{(2)}\big] \tag{7.350}$$

$$f_\pm = \frac{1}{2}\big[\epsilon_1 e^{-i\phi} \xi_{1,\pm}^{(2)} - \epsilon_2 e^{i\phi} \xi_{2,\pm}^{(2)}\big]. \tag{A.351}$$

$$\frac{\mathrm{d}\rho_s}{\mathrm{d}s} = -\frac{i}{\hbar}\big[\mathrm{H}_s, \rho_s\big] + \mathfrak{D}(\rho_s); \tag{352a}$$

$$\mathfrak{D}_d(\rho) = \gamma_d \sum_{k=1}^{3} \big[\sigma^k \rho \sigma^k - \rho\big], \tag{353}$$

$$\mathfrak{D}_r(\rho) = \gamma_1 \big[\sigma^+ \rho \sigma^- - \frac{1}{2}\big\{\mathbb{P}^{(1)}, \rho\big\}\big] + \gamma_z \big[\mathrm{Z}\rho\mathrm{Z} - \rho\big], \tag{4.354}$$

$$\mathrm{L}_1 = \sqrt{\frac{\lambda_1}{\lambda}}\sigma^-, \quad \mathrm{L}_2 = \sqrt{\frac{\lambda_2}{\lambda}}\sigma^+, \quad \mathrm{L}_3 = \sqrt{\frac{\lambda_3}{\lambda}}\mathrm{Z}; \tag{A.355}$$

$$\mathfrak{D}(\rho) = \bigoplus_{k=1}^{n} \mathfrak{D}^{(k)}(\rho), \tag{356a}$$

$$\mathcal{H}(\rho, \sigma) = \frac{1}{\sqrt{2}} \sqrt{\sum_{k=1}^{N} \big(\sqrt{\rho_{kk}} - \sqrt{\sigma_{kk}}\big)^2}, \tag{357}$$

$$\mathrm{H}(\theta, \phi) = \frac{\theta\hbar}{2}\mathrm{R}_{xy}(\phi) \tag{1.358}$$

$$\xi_{k,+} = \int_0^1 \mathrm{dW}_{k,s} \cos(s\theta), \quad \xi_{k,-} = \int_0^1 \mathrm{dW}_{k,s} \sin(s\theta), \tag{A.359}$$

$$\mathbb{E}\big[\xi_{k,\pm}^2\big] = \frac{1}{2}\big[1 \pm \frac{\sin(2\theta)}{2\theta}\big], \tag{360a}$$

$$\mathbb{E}\big[\xi_{k,+}\xi_{j,-}\big] = \frac{1 - \cos(2\theta)}{4\theta}\delta_{kj}; \tag{361}$$

$$\xi_{k,w} = \int_0^1 \mathrm{dW}_{k,s}, \tag{5.362}$$

$$\mathcal{E}(\rho) = \sum_i \mathrm{K}_i \rho \mathrm{K}_i^\dagger, \tag{A.363}$$

$$\Delta S(T, \Delta H) = \mu_0 \int_{H_i}^{H_f} \left(\frac{\partial M(T, H)}{\partial T}\right)_H dH, \tag{364a}$$

$$H_h = -\sum_{i \neq j} J_{ij} \mathbf{e}_i \cdot \mathbf{e}_j, \tag{365}$$

$$S_{ph}(\omega, T) = k_B[(n+1)\ln(n+1) - n\ln n], \tag{2.366}$$

$$\mathcal{H} = \sum_{i=1}^{N} \frac{|\boldsymbol{p}_i|^2}{2m_i} + \sum_{i,j,i\neq 1}^{N} V(r_{ij}) + \mathcal{H}_{mag} \tag{A.367}$$

$$\mathcal{H}_{mag} = -\sum_{i,j,i\neq 1}^{N} \boldsymbol{J}(r_{ij})\boldsymbol{S}_i.\boldsymbol{S}_j + \mathcal{H}_{ext}. \tag{368a}$$

$$\frac{dr_i}{dt} = \frac{p_i}{m_i}, \tag{369}$$

$$\frac{dp_i}{dt} = \sum_{i,j,i\neq 1}^{N} \left[\frac{dV(r_{ij})}{dr_{ij}} + \frac{dJ(r_{ij})}{dr_{ij}}\boldsymbol{s}_i.\boldsymbol{s}_j\right]\boldsymbol{e}_i, \tag{6.370}$$

$$\frac{d\boldsymbol{s}_i}{dt} = \boldsymbol{\omega}_i \times \boldsymbol{s}_i, \tag{A.371}$$

$$\boldsymbol{\omega}_i = -\frac{1}{\hbar}\frac{\partial \mathcal{H}_{mag}}{\partial \boldsymbol{s}_i}. \tag{372a}$$

$$S_{c,n} = S_n - S_n^{\text{Slater}} = \frac{1}{1-n}\ln\left(\text{Tr}\left(\frac{C^n}{N_p}\right)\right). \tag{373}$$

$$s_c = 1 - \frac{\langle x\rangle}{2N_p}. \tag{3.374}$$

$$S_c = -\log(s_c) \approx 4\gamma. \tag{A.375}$$

$$S_P = -\log(P(x=0)) \approx 2\gamma N_p. \tag{376a}$$

$$\hat{U} = \exp(i\vec{c}^{\dagger}A\vec{c}), \tag{377}$$

$$C_{ii'} = N_p \sum_{j,k,l,\ldots} \Phi(i,j,k,l,...)\Phi(i',j,k,l,...)^*, \tag{7.378}$$

$$S_n = \frac{1}{1-n}\ln(\text{Tr}(\rho_1^n)), \tag{A.379}$$

$$\langle x\rangle = \sum_{i,j=1}^{N_s} p_i p_j x_{ij}, \tag{380a}$$

$$s_c \approx 1 - 4\gamma. \tag{381}$$

$$|\Phi\rangle = \sum_{k=1}^{k_{max}} a_{\{n_{B_i}\}_k}|\{n_{B_i}\}_k\rangle \tag{4.382}$$

$$S_{P_{\mathcal{B}}} = -\ln\sum_k P_k^2, \tag{A.383}$$

$$S_P = \min_{\mathcal{B}} S_{P_{\mathcal{B}}}, \tag{384a}$$

$$\mathcal{C} = e^{S_P}. \tag{385}$$

$$C_{ij} = \langle \Phi | \hat{c}_j^\dagger \hat{c}_i | \Phi \rangle, \tag{1.386}$$

$$S_c^p = -\ln \frac{\sum_i \lambda_i^2}{N_p}, \tag{A.387}$$

$$S_c^h = -\ln \frac{\sum_i \tilde{\lambda}_i^2}{N_o - N_p}, \tag{388a}$$

$$S_c = \max\left\{S_c^p, S_c^h\right\}. \tag{389}$$

$$S_P \geq S_c. \tag{5.390}$$

$$S_P \sim \alpha N_i S_c, \tag{A.391}$$

$$\hat{H} = t \sum_{\langle i,j \rangle, \sigma} \left(\hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} + \text{h.c.} \right) + U \sum_{\langle i,j \rangle, \sigma} \hat{n}_{i\sigma} \hat{n}_{i\bar{\sigma}}, \tag{392a}$$

$$\hat{H} = \sum_{\langle i,j \rangle} \left(t \hat{c}_i^\dagger \hat{c}_j + \text{h.c.} + V \hat{n}_i \hat{n}_j \right), \tag{393}$$

$$P(\delta, n) = \sum_{\delta - \text{typical}} |k_1\rangle\langle k_1| \otimes |k_2\rangle\langle k_2| \ldots |k_n\rangle\langle k_n|, \tag{2.394}$$

$$||P(\delta, n)|\Psi\rangle|| = 1 - \epsilon, \tag{A.395}$$

$$\ln(\dim \mathcal{H}) \geq n S_P, \tag{396a}$$

$$|\{n_{B_i}\}_{k_1}\rangle \otimes |\{n_{B_i}\}_{k_2}\rangle \otimes \ldots \otimes |\{n_{B_i},\}_{k_n}\rangle, \tag{397}$$

$$|\psi\rangle = U|\psi_0\rangle = \sum_{k,j=1}^{Q} U_{jk} a_k |j\rangle. \tag{6.398}$$

$$\langle U_{jk} U_{jm} U_{jl}^* U_{jn}^* \rangle = \frac{1}{Q^2} \left(\delta_{kl} \delta_{mn} + \delta_{kn} \delta_{ml} \right) \tag{A.399}$$

$$\sum_{j=1}^{Q} \langle P_j^2 \rangle = \frac{2}{Q} \tag{400a}$$

$$S_P = -N_o \ln\left[\nu^\nu (1-\nu)^{1-\nu} \right], \tag{401}$$

$$\begin{aligned} \alpha_g &= 1 + \frac{(1-\nu)\ln(1-\nu)}{\nu \ln(\nu)}, \qquad 0 \leq \nu \leq \frac{1}{2} \\ \alpha_g &= 1 + \frac{\nu \ln(\nu)}{(1-\nu)\ln(1-\nu)}, \qquad \frac{1}{2} < \nu \leq 1 \end{aligned} \tag{3.402}$$

$$\mathcal{L} = \bar{\psi}\big(i\gamma^\mu \partial_\mu - e\gamma^\mu A_\mu - m\big)\psi - \frac{1}{4} F_{\mu\nu} F^{\mu\nu}, \tag{A.403}$$

$$H = \int dx \; : \left[\frac{\Pi^2}{2} + \frac{(\partial_x \phi)^2}{2} + \frac{e^2 \phi^2}{2\pi} - \frac{b\, m\, e \cos(\sqrt{4\pi}\phi - \theta)}{2\pi^{3/2}} \right] :, \tag{404a}$$

$$H = \chi \sum_x \left[\frac{\pi_x^2}{2} + \frac{(\phi_x - \phi_{x-1})^2}{2} + \frac{\mu^2 \phi_x^2}{2} - \lambda \cos(\beta \phi_x - \theta) \right], \tag{405}$$

$$\partial_x E = -\partial_x^2 A_0 = e\rho, \tag{7.406}$$

$$H = \int dx \; : \left[\frac{\Pi^2}{2} + \frac{(\partial_x \phi)^2}{2} + \frac{M^2(\phi + \theta/\beta)^2}{2} - u \cos(\beta \phi) \right] :, \tag{A.407}$$

$$M = \frac{e}{\sqrt{\pi}} \frac{1}{\sqrt{1 + g/\pi}}, \quad u = \frac{\exp(\gamma)}{2\pi^{3/2}} m\, e, \quad \beta = \sqrt{\frac{4\pi}{1 + g/\pi}}, \tag{408a}$$

$$\overline{\psi} \gamma^\mu \psi \; = \; -\frac{\beta}{2\pi} \epsilon^{\mu\nu} \partial_\nu \phi. \tag{409}$$

$$H = \int dx \; : \left[\frac{\Pi^2}{2} + \frac{(\partial_x \phi)^2}{2} + \frac{M^2 \phi^2}{2} - u \cos(\beta \phi - \theta) \right] :, \tag{4.410}$$

$$-C\ddot{\phi}_i - \frac{1}{L}\phi_i + I_c \sin(\Phi_{\text{ext}} - \phi_i) + \frac{1}{L'}(\phi_{i-1} - 2\phi_i + \phi_{i+1}) = 0. \tag{A.411}$$

$$\mathcal{L} = \sum_i \left[\frac{C\dot{\phi}_i^2}{2} - \frac{(\phi_i - \phi_{i-1})^2}{2L'} - \frac{\phi_i^2}{2L} - I_c \cos(\phi_i - \Phi_{\text{ext}}) \right]. \tag{412a}$$

$$H = \sum_i \left[4E_C \pi_i^2 + \frac{E_{L'}(\phi_i - \phi_{i-1})^2}{2} + \frac{E_L \phi_i^2}{2} + E_J \cos(\phi_i - \Phi_{\text{ext}}) \right], \tag{413}$$

$$H = \chi \sum_i \left[\frac{\pi_i^2}{2} + \frac{(\phi_i - \phi_{i-1})^2}{2} + \frac{\mu^2 \phi_i^2}{2} - \lambda \cos(\beta \phi_i - \theta) \right], \tag{1.414}$$

$$\Delta(a) = \int_{-\pi}^{\pi} \frac{d^2k}{(2\pi)^2} \frac{1}{4 \sum_\mu \sin[2](\frac{k_\mu}{2}) + (Ma)^2} \tag{A.415}$$

$$H = \sum_j \sum_k \omega_{k,j} \Psi_{k,j}^\dagger \Psi_{k,j} + \text{interactions}, \tag{416a}$$

$$H_{\text{qubit}} = \frac{\Delta}{2} \sigma_z + g(t) \sigma_x (a_L + a_L^\dagger), \tag{417}$$

$$H = \chi \sum_{i,j} \left[\frac{\pi_{i,j}^2}{2} + \frac{(\phi_{i.j} - \phi_{i-1,j})^2 + (\phi_{i,j} - \phi_{i,j-1})^2}{2} + \frac{\mu^2 \phi_{i,j}^2}{2} - \lambda \cos(\beta \phi_{i,j} - \theta) \right], \tag{5.418}$$

$$H \approx \chi \sum_{i,j} \left[\frac{\pi_{i,j}^2}{2} + \frac{(\phi_{i.j} - \phi_{i-1,j})^2 + (\phi_{i,j} - \phi_{i,j-1})^2}{2} + \frac{m^2 \phi_{i,j}^2}{2} + \frac{\tilde{\lambda}}{4!} \phi_{i,j}^4 \right], \tag{A.419}$$

$$B_n \equiv \int_{-\pi}^{\pi} dp\, c_p\, e^{ipn} B(p). \tag{420a}$$

$$](10,0)to[barrier, i <= \tag{421}$$

$$](8,-2)node[ground]to(6,-2)to[C, i <= \tag{2.422}$$

$$\partial_t \mathbf{P}(\omega, \vec{v}) = \mathbf{H}(\omega, \vec{v}) \times \mathbf{P}(\omega, \vec{v}), \tag{A.423}$$

$$\mathbf{J}^\rho = \int d\vec{v}\, v^\rho(\vec{v}) \int_{-\infty}^{+\infty} d\omega\, F(\omega, \vec{v}) \mathbf{P}(\omega, \vec{v}) \tag{424a}$$

$$\mathsf{Q} = \frac{\mathsf{T}}{H_3 - \Omega}, \tag{425}$$

$$\partial_t \mathbf{P}^{\mathrm{II}} = \mathbf{H}^{\mathrm{II}} \times \mathbf{P}^{\mathrm{II}}, \tag{6.426}$$

$$\mathbf{H}^{\mathrm{II}} = \mathbf{X}^{\mathrm{II}} + u\mathbf{Y}^{\mathrm{II}} - \Omega_r \hat{\mathbf{e}}_3^{\mathrm{II}}. \tag{A.427}$$

$$\partial_t \mathbf{P}^{\mathrm{III}} = \mathbf{H}^{\mathrm{III}} \times \mathbf{P}^{\mathrm{III}}, \tag{428a}$$

$$\Theta(t) = 2 \arctan[e^{\Omega_i (t - t_0)}], \tag{429}$$

$$\partial_t \mathbf{P}_{\omega,\vec{v}} = \mathbf{H}_{\omega,\vec{v}} \times \mathbf{P}_{\omega,\vec{v}}, \tag{3.430}$$

$$g(v^z, \alpha) = \frac{20}{\sqrt{\pi}} \left[\sigma_\nu^{-1} e^{-(\frac{1-v^z}{\sigma_\nu})^2} - \alpha \sigma_{\bar{\nu}}^{-1} e^{-(\frac{1-v^z}{\sigma_{\bar{\nu}}})^2} \right], \tag{A.431}$$

$$\mathsf{Q}(u) = \frac{\mathsf{A} + \mathsf{B} h(u)}{\mathsf{C} + \mathsf{D} h(u)}, \tag{432a}$$

$$\mathsf{D}\,\mathsf{Q}(u) = \frac{\mathsf{A} - |\mathsf{C}/\mathsf{D}|\mathsf{B}}{|\mathsf{C}/\mathsf{D}| + h(u)} + \mathsf{B} \tag{433}$$

$$\mathsf{Q}(u) = \mathsf{O} + \mathsf{R} \frac{\mathsf{C}^* + \mathsf{D}^* h(u)}{\mathsf{C} + \mathsf{D} h(u)}. \tag{7.434}$$

$$\mathrm{Im}(\mathsf{A}\,\mathsf{B}^*) = 0. \tag{A.435}$$

$$\rho(r) = \begin{cases} \frac{\rho_0}{(\frac{r}{r_s})(1 + \frac{r}{r_s})^2} & \text{if } r < R_{\mathrm{trunc}} \\ 0 & \text{if } r > R_{\mathrm{trunc}} \end{cases}, \tag{436a}$$

$$\rho_0 = \rho_{\mathrm{crit}}(z)\, \frac{200}{3}\, \frac{c_{200}}{\ln(1 + c_{200}) - \left(\frac{c_{200}}{1 + c_{200}} \right)}, \tag{437}$$

$$c_{200}(M_{200}, z) = 5.71\, (1+z)^{-0.47} \left(\frac{M_{200}}{2 \times 10^{12}\, h^{-1} M_\odot} \right)^{-0.084}. \tag{4.438}$$

$$\widetilde{X}(\hat{\mathbf{n}}) = X(\hat{\mathbf{n}} + \boldsymbol{\alpha}(\hat{\mathbf{n}})), \tag{A.439}$$

$$\kappa(\hat{\mathbf{n}}) = -\frac{1}{2}\nabla^2_{\hat{\mathbf{n}}}\phi(\hat{\mathbf{n}}). \tag{440a}$$

$$\kappa_{\rm cl}(r) = \frac{\Sigma_{\rm cl}(r)}{\Sigma_{\rm crit}(z)}, \tag{441}$$

$$\Sigma_{\rm cl}(r) = 2\int_r^{R_{\rm trunc}} \frac{x\rho(x)}{\sqrt{x^2 - R^2}}{\rm d}x \ , \tag{1.442}$$

$$\Sigma_{\rm crit}(z) = \frac{c^2}{4\pi G}\frac{d_{A,\rm CMB}}{d_{A,\rm cl}d_{A,\text{cl-CMB}}} \ , \tag{A.443}$$

$$\kappa_{cl}(x) = \frac{2\rho_s r_s}{\Sigma_{crit}(z)}g(x),, \tag{444a}$$

$$\kappa_0\theta_s^2 \propto \frac{M_{200}}{\Sigma_{\rm crit}(z)d_A^2(z)} \tag{445}$$

$$N(L) = C_L^{\kappa\kappa} + N_L^{(0)} + N_L^{(1)}, \tag{5.446}$$

$$\frac{2L+1}{4\pi}\frac{\hat{\kappa}(L)\kappa_t(L)}{N(L)}, \tag{A.447}$$

$$\kappa_{cl} = \frac{2\rho_s r_s}{\Sigma_{crit}(z)}g(x), \text{ where } x = \frac{r}{r_s} = \frac{\theta}{\theta_s}. \tag{448a}$$

$$\begin{aligned} {}_1\hat{g}(\hat{n}) = &\left(\sum_{l_1m_1} F_{l_1}T_{l_1m_1}\ {}_sY_{l_1m_1}(\hat{n})\right) \\ &\times \left(\sum_{l_2m_2} G_{l_1}T_{l_2m_2}\ {}_tY_{l_2m_2}(\hat{n})\right). \end{aligned} \tag{449}$$

$$F_l = \frac{1}{C_l + N_l}, \quad G_l = -\sqrt{l(l+1)}\frac{C_l}{C_l + N_l}, \tag{2.450}$$

$$\begin{aligned} G_{lm} &= \int d^2n_2 G(\hat{n}, \hat{n}_2)Y^*_{lm}(\hat{n}'_2) \\ F_{lm} &= \int d^2n_1 F(\hat{n}, \hat{n}_1)Y^*_{lm}(\hat{n}'_1). \end{aligned} \tag{A.451}$$

$$\begin{aligned} F(\hat{n}, \hat{n}_1) &= \sum_{l_1m_1} F_{l_1}\ {}_sY_{l_1m_1}(\hat{n})\ {}_0Y^*_{l_1m_1}(\hat{n}_1), \\ G(\hat{n}, \hat{n}_2) &= \sum_{l_2m_2} G_{l_2}\ {}_tY_{l_2m_2}(\hat{n})\ {}_0Y^*_{l_2m_2}(\hat{n}_2). \end{aligned} \tag{452a}$$

$$\tilde{\mu}_c = \frac{\tilde{\mu}_{uc} + \tilde{\mu}_{dc}}{2} \tag{453}$$

$$\delta\tilde{\mu}_c = \frac{\tilde{\mu}_{uc} - \tilde{\mu}_{dc}}{2}. \tag{6.454}$$

$$\epsilon_c^{\pm} = \bar{E}_c^{\pm}\sqrt{1 + \left[g(\vec{p})\bar{\Delta}(1-\delta_{bc})/\bar{E}_c^{\pm}\right]^2} \tag{A.455}$$

$$\bar{E}_c^{\pm} = E \pm \tilde{\mu}_c. \tag{456a}$$

$$M(\vec{p}) = m_c + g(\vec{p})\bar{\sigma}. \tag{457}$$

$$P = P_q + P_{lep}\ , \tag{3.458}$$

$$\mu_{Q_q} = \mu_{uc} - \mu_{dc} = -\mu_e + \mu_{\nu_e} \tag{A.459}$$

$$B(\mu_B) = B_0\, f_<(\mu_B) \tag{460a}$$

$$f_<(\mu_B) = \frac{1}{2}\left[1 - \tanh\left(\frac{\mu_B - \mu_<}{\Gamma_<}\right)\right], \tag{461}$$

$$\varepsilon = -P^* + T\ s + G \tag{7.462}$$

$$G = n_B^*\ \mu_B + (n_{\nu_e} + n_e)\ \mu_{\nu_e} \tag{A.463}$$

$$g_{\tau B}(n) = g_{\tau B}(n_0)\ a_\tau\ \frac{1 + b_\tau(\frac{n}{n_0} + d_\tau)^2}{1 + c_\tau(\frac{n}{n_0} + d_\tau)^2}\,, \tag{464a}$$

$$g_{\rho B}(n) = g_{\rho B}(n_0)\exp\left[-a_\rho\left(\frac{n}{n_0} - 1\right)\right], \tag{465}$$

$$f_{B\mp}(p) = \frac{1}{\exp\left[\frac{E_B^*(p) \mp \mu_B^*}{T}\right] + 1}, \quad E_B^*(p) = \sqrt{p^2 + m_B^{*2}}\,, \tag{4.466}$$

$$\mu_B^* = \mu_B - g_{\omega B}(n)\bar{\omega} - g_{\rho B}(n)\bar{\rho}I_{3B} - \widetilde{R}\,, \tag{A.467}$$

$$P_L = \sum_L \frac{\gamma_L}{3}\int \frac{d^3p}{(2\pi)^3}\frac{p^2}{E_L}[f_{L-}(p) + f_{L+}(p)]\,, \tag{468a}$$

$$\varepsilon = -P + TS + \sum_j \mu_j\, n_j\,, \tag{469}$$

$$\Omega_{reg}^{MFA} = \Omega^{MFA} - \Omega_0^{MFA}\,. \tag{1.470}$$

$$\begin{aligned} \frac{dP(r)}{dr} &= \frac{G(\varepsilon(r)) + P(r))(M(r) + 4\pi r^3 P(r))}{r(r - 2GM(r))}, \\ \frac{dM(r)}{dr} &= 4\pi r^2 \varepsilon(r)\,, \end{aligned} \tag{A.471}$$

$$|B(q, 5d(q,\beta))| \leq (c(R \cup \{q\}))^3 \cdot |B(q, \frac{5}{8}d(q,\beta))| \tag{472a}$$

$$\{q \in \text{Children}(p) \mid l(q) = i\} \cup \{p\} \leq (c_m(R))^4, \tag{473}$$

$$|\bar{B}(q, \frac{4}{3} \cdot d(q, p_n))| \geq (1 + \frac{1}{c(A)^2})^n \cdot |\bar{B}(q, \frac{1}{3} \cdot d(q, p_1))|. \tag{5.474}$$

$$\eta(i) = \min\{t \in L(\mathcal{T}(W), p) \mid t > i\}. \tag{A.475}$$

$$O\Big((c_m(R))^8 \cdot \max_{y=1,\ldots,|R|-1} L(\mathcal{T}(W_y), p_y) \cdot |R|\Big), \tag{476a}$$

$$\max_{y=2,\ldots,|R|} |L(\mathcal{T}(W_{y-1}), p_y)| \leq H(\mathcal{T}(R)) \leq 1 + \log_2 \Delta(R). \tag{477}$$

$$|\bar{B}(q, \frac{4}{3} d(q, p_n))| \geq (1 + \frac{1}{c(R)^2})^n \cdot |\bar{B}(q, \frac{1}{3} d(q, p_1))| \tag{2.478}$$

$$|A| \geq \frac{|\bar{B}(q, \frac{4}{3} \cdot d(q, p_n))|}{|\bar{B}(q, \frac{1}{3} \cdot d(q, p_1))|} \geq (1 + \frac{1}{c(A)^2})^{|S|} \tag{A.479}$$

$$O\Big(\log_2(k) \cdot ((c_m(R))^{10} \cdot |L(q, \mathcal{T}(R))| + |\bar{B}(q, 5d_k(q, R))|)\Big), \tag{480a}$$

$$I_i^1(u_i^t | l_i^t) = \sum_{u_i^{t-1}, y_i^{t-1}, a_{-i}^{t-1}} P^1(u_i^t | x_i^{t-1}, u_i^{t-1}, y_i^{t-1}, a^{t-1}) P^1(u_i^{t-1}, y_i^{t-1}, a_{-i}^{t-1} | l_i^t) \tag{481}$$

$$I_i^2(u_i^t | l_i^t) = \sum_{u_i^{t-1}, y_i^{t-1}, a_{-i}^{t-1}} P^2(u_i^t | x_i^{t-1}, u_i^{t-1}, y_i^{t-1}, a^{t-1}) P^2(u_i^{t-1}, y_i^{t-1}, a_{-i}^{t-1} | l_i^t). \tag{6.482}$$

$$P^1(u_i^{t-1}, y_i^{t-1}, a_{-i}^{t-1} | l_i^t) = \sum_{h_{-i}^{t-1}} \pi_{-i}(a_{-i}^{t-1} | h_{-i}^{t-1}) P^1(u_i^{t-1} y_i^{t-1}, h_{-i}^{t-1} | l_i^t) \tag{A.483}$$

$$P^2(u_i^{t-1}, y_i^{t-1}, a_{-i}^{t-1} | l^t) = \sum_{h_{-i}^{t-1}} \pi_{-i}(a_{-i}^{t-1} | h_{-i}^{t-1}) P^2(u_i^{t-1} y_i^{t-1}, h_{-i}^{t-1} | l_i^t) \tag{484a}$$

$$I_i^1(u_i^t | l_i^t) = I_i^2(u_i^t | l_i^t) \tag{485}$$

$$P(u_i^t | x_i^{t-1}, u_i^{t-1}, y_i^{t-1}, a^{t-1}) = P(u_i^t | x_i^{t-1}, u_i^{t-1}, y_i^{t-1}, a_i^{t-1}) \tag{3.486}$$

$$\forall i \in N : \exists! I_i^*(u_i^t | l_i^t) \in \Psi_i : \forall \pi \in \Pi \left(I_i(u_i^t | l_i^t, \pi) = I_i^*(u_i^t | l_i^t)\right) \tag{A.487}$$

$$\sum_{x_i^{t+1}} \left|P^1(h_i^{t+1} | h_i^t, a_i^t) - P^2(h_i^{t+1} | h_i^t, a_i^t)\right| \leq \sum_{l_i^t, u_i^t} P(l_i^t | h_i^t) \left|I^1(u_i^t | l_i^t) - I^2(u_i^t | l_i^t)\right| \quad \forall h_i^t, a_i^t \tag{488a}$$

$$\sum_{h_i^{t+1}} \left(P^1(h_i^{t+1} | h_i^t, a_i^t) - P^2(h_i^{t+1} | h_i^t, a_i^t)\right) \leq \sum_{l_i^t, u_i^t} P(l_i^t | h_i^t) \left|I^1(u_i^t | l_i^t) - I^2(u_i^t | l_i^t)\right| \leq \xi \quad \forall h_i^t, a_i^t. \tag{489}$$

$$\left|Q_{M_i^1}^{\pi_i}(h_i^t, a_i^t) - Q_{M_i^2}^{\pi_i}(h_i^t, a_i^t)\right| \leq \sum_{k=t}^{H} \bar{R}(H-k)\xi = \bar{R}\frac{(H-t)(H-t+1)}{2}\xi. \tag{7.490}$$

$$Q_{M_i^2}^{\pi^*}(h_i^t, \bar{a}_i^t) < Q_{M_i^2}^{\pi^*}(h_i^t, \hat{a}_i^t) \tag{A.491}$$

$$Q_{M_i^1}^{\pi^*}(h_i^t, \bar{a}_i^t) - \Delta \leq Q_{M_i^2}^{\pi^*}(h_i^t, \bar{a}_i^t) < Q_{M_i^2}^{\pi^*}(h_i^t, \hat{a}_i^t) \leq Q_{M_i^1}^{\pi^*}(h_i^t, \hat{a}_i^t) + \Delta. \tag{492a}$$

$$Q_{M_i^1}^{\pi^*}(h_i^t, \bar{a}_i^t) - Q_{M_i^1}^{\pi^*}(h_i^t, \hat{a}_i^t) < 2\Delta, \tag{493}$$

$$Q_{M_i^1}^{\pi^*}(h_i^t, \bar{a}_i^t) - Q_{M_i^1}^{\pi^*}(h_i^t, a_i^t) > 2\Delta \quad \forall h_i^t, a_i^t \tag{4.494}$$

$$\hat{S} = 1_{[v,1]} \circ a \circ \hat{Y} \circ X, \tag{A.495}$$

$$\begin{aligned} BA(b_1) &= \frac{1}{2}(P(\hat{Y}=0|S=0) + P(\hat{Y}=1|S=1)) \\ &= \frac{1}{2}(1 + P(\hat{Y}=1|S=1) - P(\hat{Y}=1|S=0)) \end{aligned} \tag{496a}$$

$$BA(b_2) = \frac{1}{2}(1 + P(\hat{Y}=1|S=0) - P(\hat{Y}=1|S=1)) \tag{497}$$

$$\begin{aligned} P(\hat{S}=0|S=0) =& P(\hat{S}=0|S=0Y=0)P(Y=0|S=0) \\ &+ P(\hat{S}=0|S=0Y=1)P(Y=1|S=0) \end{aligned} \tag{1.498}$$

$$\begin{aligned} P(\hat{S}=1|S=1) =& P(\hat{S}=1|S=1Y=0)P(Y=0|S=1) \\ &+ P(\hat{S}=1|S=1Y=1)P(Y=1|S=1) \end{aligned} \tag{A.499}$$

$$\begin{aligned} BA(b_1) &= \frac{1}{2}(P(\hat{Y}=0|S=0) + P(\hat{Y}=1|S=1)) \\ &= \frac{1}{2}(1 + P(\hat{Y}=1|S=1) - P(\hat{Y}=1|S=0)) \\ BA(b_2) &= \frac{1}{2}(1 + P(\hat{Y}=1|S=0) - P(\hat{Y}=1|S=1)) \end{aligned} \tag{500a}$$

$$formula, \tag{501}$$

$$\boldsymbol{\theta}^* = \underset{\boldsymbol{\theta}}{\operatorname{argmin}} \left(-G(\boldsymbol{\theta}) + \lambda \mathcal{R}(\boldsymbol{\theta})\right). \tag{5.502}$$

$$\begin{aligned} \mathcal{D}(\mathcal{E}, \boldsymbol{\theta}) &\doteq \{\nabla \cdot \mathbf{f}_k\}_{k=1}^{N_e}, \\ \mathcal{A}(\mathcal{E}, \boldsymbol{\theta}) &\doteq \left\{|\det(\mathtt{J}(e_k))|\right\}_{k=1}^{N_e}, \end{aligned} \tag{A.503}$$

$$|\mathtt{J}_{t,t+\Delta t}| \doteq \left|\det\left(\frac{d\mathbf{x}(t+\Delta t)}{d\mathbf{x}(t)}\right)\right| = \left(\frac{1-th_z}{1-(t+\Delta t)h_z}\right)^2. \tag{504a}$$

$$|\mathtt{J}_{t,t+\Delta t}| = 1 + \left.\frac{d|\mathtt{J}_{t,t+\Delta t}|}{d\Delta t}\right|_{\Delta t=0} \Delta t + \cdots \tag{505}$$

$$\left.\frac{d|\mathtt{J}_{t,t+\Delta t}|}{d\Delta t}\right|_{\Delta t=0} = \frac{2h_z}{1-th_z}. \tag{2.506}$$

$$\mathbf{x}^h(t) \sim \mathtt{R}(\boldsymbol{\omega} t)\, \mathbf{x}^h(0), \tag{A.507}$$

$$\mathbf{x}^h(t+\Delta t) \sim \mathtt{R}(\boldsymbol{\omega}\Delta t)\,\mathbf{x}^h(t). \tag{508a}$$

$$\det\left(\frac{d\mathbf{x}(t+\Delta t)}{d\mathbf{x}(t)}\right) = \left(\mathbf{r}_3^\top(\boldsymbol{\omega}\Delta t)\mathbf{x}^h(t)\right)^{-3}, \tag{509}$$

$$\left.\frac{d|\mathtt{J}_{t,t+\Delta t}|}{d\Delta t}\right|_{\Delta t=0} = 3\mathbf{x}^{h\top}(t)\boldsymbol{\omega}^\wedge\mathbf{e}_3. \tag{6.510}$$

$$\begin{aligned}\mathcal{R}_{\mathbf{x}(0)} &= \int_0^1 \left.\frac{d|\mathtt{J}_{t,t+\Delta t}|}{d\Delta t}\right|_{\Delta t=0} dt \\ &= 3\omega_x\int_0^1 y(t)dt - 3\omega_y\int_0^1 x(t)dt.\end{aligned} \tag{A.511}$$

$$\mathbf{x}'_k = \mathbf{x}_k - \mathbf{v}(\mathbf{x})t_k. \tag{512a}$$

$$h_z = \frac{v_z}{Z}, \tag{513}$$

$$\mathbf{x}_k^{h\prime} \sim \mathtt{H}^{-1}(t_k;\boldsymbol{\theta})\,\mathbf{x}_k^h, \tag{3.514}$$

$$\mathbf{x}^h(t) \sim \mathtt{H}(t;\boldsymbol{\theta})\,\mathbf{x}^h(0). \tag{A.515}$$

$$\mathbf{x}^h(t+\Delta t) \sim \underbrace{\mathtt{H}(t+\Delta t;\boldsymbol{\theta})\,\mathtt{H}^{-1}(t;\boldsymbol{\theta})}_{\mathtt{H}_{t,t+\Delta t}}\mathbf{x}^h(t). \tag{516a}$$

$$\det\big(\mathtt{J}_{t,t+\Delta t}\big) = \frac{\det(\mathtt{H}_{t,t+\Delta t})}{(\mathbf{e}_3^\top\mathtt{H}_{t,t+\Delta t}\,\mathbf{x}^h(t))^3}, \tag{517}$$

$$\det\big(\mathtt{J}_{t,t+\Delta t}\big) = \frac{1}{(\mathbf{e}_3^\top\mathtt{R}(\boldsymbol{\omega}\Delta t)\,\mathbf{x}^h)^3}, \tag{7.518}$$

$$\mathtt{H}^S(t;\boldsymbol{\theta}) \doteq \begin{pmatrix}\beta(t;s)\mathtt{R}(t\omega_z) & t\mathbf{v}\\ \mathbf{0}^\top & 1\end{pmatrix}. \tag{A.519}$$

$$\mathtt{H}^A(t;\boldsymbol{\theta}) \doteq \begin{pmatrix}\mathtt{A}(t;\boldsymbol{\theta}) & t\mathbf{b}\\ \mathbf{0}^\top & 1\end{pmatrix}, \tag{520a}$$

$$X = V\Sigma V^T \tag{521}$$

$$X_j = V_j\Sigma_j V_j^T \tag{4.522}$$

$$P(w_1^T) = \prod_{t=1}^{T} P(w_t|w_1^{t-1}) \tag{A.523}$$

$$\prod_{t=1}^{T} P(w_t|w_1^{t-1}) \approx \prod_{t=1}^{T} P(w_t|w_{t-n+1}^{t-1}) \tag{524a}$$

$$\sum_{-c\le j\le c, j\neq 0} \log p(w_i|w_{i+j}) \tag{525}$$

$$\sum_{-c\leq j\leq c, j\neq 0} \log p(w_{i+j}|w_i) \tag{1.526}$$

$$p(w_c|w_o) = \frac{\exp\left(v_{w_c}^{'T} v_{w_o}\right)}{\sum\limits_{w=1}^{W} exp(v_w^{'T} v_{w_o})} \tag{A.527}$$

$$p(w_c|w_o) = \log\sigma(v_{w_c}^{'T} v_{w_o}) + \sum_{i=1}^{k}\left[\log\sigma(-v_{w_i}^{'T} v_{w_o})\right] \tag{528a}$$

$$P(w_i) = 1 - \sqrt{\frac{t}{f(w_i)}} \tag{529}$$

$$f(x) = \begin{cases} (x/x_{max})^{\alpha} & \text{if } x < x_{max} \\ 1 & \text{otherwise} \end{cases} \tag{5.530}$$

$$\sum_{t=1}^{T}\left[\sum_{c\in C_t} \ell(v_{w_t}^{T} v_{w_c}^{'}) + \sum_{i=1}^{k} \ell(-v_{w_t}^{T} v_{w_i}^{'})\right] \tag{A.531}$$

$$v_w = \sum_{m\in\mathcal{M}_w} z_m \tag{532a}$$

$$p(t_1, t_2, ..., t_N) = \prod_{i=1}^{N} p(t_i|t_1, ..., t_{i-1}) \tag{533}$$

$$p(t_1, t_2, ..., t_N) = \prod_{i=1}^{N} p(t_i|t_{i+1}, t_{i+2}, ..., t_N) \tag{2.534}$$

$$R_i = \{h_{i,j}^{LM} | j = 0, ..., L\} \tag{A.535}$$

$$\int_{\mathbb{R}} K(x)dx = 1, \quad \|K\|_{\infty} < \infty, \quad \text{supp}(K) \subset [-1, 1], \quad \int_{\mathbb{R}} K(x)x^l dx = 0, \tag{536a}$$

$$Var(\hat{\pi}_{h,n}(x^*)) \leq \frac{c\Delta_n}{T_n h} + 2\,|I| \tag{537}$$

$$|\varphi_x(\xi, t)| \leq \tilde{c}(1 + |\xi|)^{-2}. \tag{6.538}$$

$$\int_{\mathbb{R}} |\partial_y^2 p_1(z, y)|dy < \infty. \tag{A.539}$$

$$\tilde{c}_{\pi} - c_{\pi} = O(\frac{h_n}{M_n}). \tag{540a}$$

$$H^2(\pi, \tilde{\pi}) \leq \frac{Ch_n}{M_n^2}. \tag{541}$$

$$\dot{X}_t^\epsilon = \int_0^t b'(X_s^\epsilon)\dot{X}_s^\epsilon ds + \int_0^t \left[\frac{\epsilon}{M_n}\psi'_{h_n}(X_s^\epsilon)\dot{X}_s^\epsilon + \frac{1}{M_n}\psi_{h_n}(X_s^\epsilon)\right] dB_s. \tag{3.542}$$

$$Y_t^\epsilon = \exp(\frac{\epsilon}{M_n}\int_0^t \psi'_{h_n}(X_s^\epsilon)dB_s - \frac{\epsilon^2}{2M_n^2}\int_0^t (\psi'_{h_n}(X_s^\epsilon))^2 ds). \tag{A.543}$$

$$Y_t^\epsilon \leq \exp(\frac{c}{M_n} + \frac{c\,t}{h_n^2 M_n} + \frac{c\,t}{h_n^2 M_n^2}). \tag{544a}$$

$$\sup_{x_0\in\mathbb{R}} \mathbb{E}_{x_0}\left[\left|\Phi_\Delta - \frac{B_{\Delta_n}}{\sqrt{\Delta_n}}\right|^p\right] \leq \frac{c(p)}{(M_n h_n^2)^p}. \tag{545}$$

$$Var(\hat{\pi}_{h,n}(x^*)) \leq \frac{c}{T_n} + \frac{c}{T_n}\frac{\Delta_n}{h}. \tag{7.546}$$

$$\int_{\mathbb{R}} |\mathbb{K}_h(x^* - z)|dz \leq c, \tag{A.547}$$

$$\int_{\mathbb{R}} \mathbb{E}_{x_0}[(\frac{\dot{p}^\epsilon_{\Delta_n}(x_0, X^\epsilon_{\Delta_n})}{p^\epsilon_{\Delta_n}(x_0, X^\epsilon_{\Delta_n})})^2]\pi^\epsilon(x_0)dx_0 \leq \hat{c}\frac{h_n}{M_n^2}, \tag{548a}$$

$$\mathbb{P}_{x_0}(\Omega_n^c) \leq c_r n^{-r}, \tag{549}$$

$$D_u(< D.X^\epsilon_{\Delta_n}, D.X^\epsilon_{\Delta_n} >) = 2\int_0^{\Delta_n} D^2_{u,r}X^\epsilon_{\Delta_n} D_r X^\epsilon_{\Delta_n} dr \tag{4.550}$$

$$p^\epsilon_{s,\Delta_n}(z \mid x, y) = \frac{p^\epsilon_s(x,z)p^\epsilon_{\Delta_n - s}(z,y)}{p^\epsilon_{\Delta_n}(x,y)}, \tag{A.551}$$

$$\ell_{x,y,\Delta_n,h_n} \leq C\sqrt{\Delta_n}h_n e^{3\eta\frac{(y-x)^2}{\Delta_n}} + C\Delta_n e^{2\eta\frac{(y-x)^2}{\Delta_n}}. \tag{552a}$$

$$\begin{aligned} |Y^\epsilon_{\Delta_n} - 1| &\leq \frac{c\Delta_n}{M_n h_n^2}, \\ \sup_{u\in[0,\Delta_n]} |(Y^\epsilon_u)^{-1} - 1| &\leq \frac{c\Delta_n}{M_n h_n^2}. \end{aligned} \tag{553}$$

$$\begin{aligned} xb(x) &= -\eta x\mathrm{sgn}(\frac{x-x^\star}{A}) = -\eta(x-x^*)\mathrm{sgn}(\frac{x-x^\star}{A}) - \eta x^*\mathrm{sgn}(\frac{x-x^*}{A}) \\ &\leq -\eta|x-x^*| + \eta|x^*| \leq -\eta|x| + 2\eta|x^*|. \end{aligned} \tag{1.554}$$

$$a^2_{\min} \leq a^2(x). \tag{A.555}$$

$$\left\|g^{(\lfloor\beta\rfloor)}(.+t) - g^{(\lfloor\beta\rfloor)}(.)\right\|_\infty \leq \mathcal{L}|t|^{\beta-\lfloor\beta\rfloor} \qquad \forall t\in\mathbb{R}, \tag{556a}$$

$$\sup_{y\in\mathbb{R}}\int_0^\infty |g_u(x,y)|du \leq k(x), \qquad \text{for all } x\in\mathbb{R}. \tag{557}$$

$$\sup_{(a,b)\in\Sigma} \mathbb{E}_{(a,b)}[|\hat{\pi}_{h_n,n}(x^*) - \pi(x^*)|^2] \leq c(\frac{1}{n})^{\frac{2\beta}{2\beta+1}}. \tag{5.558}$$

$$k(t_j) = \mathrm{Cov}(\mathbb{K}_h(x^* - X_{t_j}), \mathbb{K}_h(x^* - X_0)) = \int_{\mathbb{R}} \int_{\mathbb{R}} \mathbb{K}_h(x^* - y)\mathbb{K}_h(x^* - z) g_{t_j}(y, z) dy\, dz \tag{A.559}$$

$$\Delta_n > (\frac{1}{n})^{\frac{1}{2\beta+1}}. \tag{560a}$$

$$H^2(\mathbb{P}, \mathbb{Q}) := \int_\Omega (\sqrt{\frac{d\mathbb{P}}{d\mu}} - \sqrt{\frac{d\mathbb{Q}}{d\mu}})^2 d\mu. \tag{561}$$

$$H_x^2(p, q) = \int_{\mathbb{R}} (\sqrt{p(x, y)} - \sqrt{q(x, y)})^2 dy. \tag{2.562}$$

$$q_{\Delta_n}(x, y)dy = \mathbb{P}(\tilde{X}_{\Delta_n} \in dy | X_0 = x). \tag{A.563}$$

$$\Omega_n := \{\omega : \, X_s^\epsilon(\omega) \notin [-h_n, h_n] \quad \forall s \in [0, \Delta_n]\}. \tag{564a}$$

$$I_1 + I_2 \leq c\frac{h_n}{M_n^2}. \tag{565}$$

$$\ell_\alpha(\mathcal{I}) = \min(|\mathcal{I} \setminus \mathcal{J}| : \mathcal{J} \subseteq \mathcal{I}, \bar{\phi}_{\mathcal{J}} = 0), \tag{6.566}$$

$$J_{\mathcal{I}} = \big\{ \bigcap_{i \in \mathcal{I}} H_i \big\} \cap \big\{ \bigcap_{j \in [n] \setminus \mathcal{I}} H_j^c \big\}, \tag{A.567}$$

$$H_{\mathcal{I}} : \big\{ \bigcap_{i \in \mathcal{I} \cap \mathcal{S}^-} H_i^- \big\} \cap \big\{ \bigcap_{i \in \mathcal{I} \cap \mathcal{S}^+} H_i^+ \big\} \tag{568a}$$

$$\mathcal{S} = (sign(p_1 - 1/2), \ldots, sign(p_n - 1/2)), \quad sign(p_i - 1/2) = \begin{cases} -1 & \text{if } p_i \leq 1/2, \\ 1 & \text{if } p_i > 1/2. \end{cases} \tag{569}$$

$$\big\{ \bigcap_{i \in \mathcal{K}^c} H_i^- \big\} \cap \big\{ \bigcap_{j \in \mathcal{K}} K_j \big\} \quad \subseteq \quad \big\{ \bigcap_{i \in \mathcal{K}^c \cap \mathcal{S}^-} H_i^- \big\} \cap \big\{ \bigcap_{j \in \mathcal{K} \cap \mathcal{S}^+} H_j^+ \big\}. \tag{3.570}$$

$$|(\mathcal{I} \cap \mathcal{S}^-) \setminus \mathcal{U}| = |\mathcal{I} \cap \mathcal{S}^- \cap \mathcal{K}| \leq \tilde{\ell}_\alpha^+(\mathcal{I}) < \ell_\alpha^+(\mathcal{I}). \tag{A.571}$$

$$y_{ij} = \gamma + \mu_i + \epsilon_{ij} \tag{572a}$$

$$f_v(\mathcal{I}) = \max_{\mathcal{K}: |\mathcal{K} \cap \mathcal{I}| = v} f(\{2p_i, i \in \mathcal{S}^- \cap \mathcal{K}^c\} \cup \{2q_i, i \in \mathcal{S}^+ \cap \mathcal{K}\}) \leq \alpha. \tag{573}$$

$$f_{v,u}(\mathcal{I}) = \max_{\substack{\mathcal{K}: |\mathcal{K} \cap \mathcal{I}| = v, \\ |\mathcal{K} \cap \mathcal{I}^c| = u}} f(\{2p_i, i \in \mathcal{S}^- \cap \mathcal{K}^c\} \cup \{2q_i, i \in \mathcal{S}^+ \cap \mathcal{K}\}) \tag{7.574}$$

$$f_{v,u}(\mathcal{I}) = \max_{\substack{k \in \{k_1, \ldots, k_2\} \\ j \in \{j_1, \ldots, j_2\}}} f(\{a_1, \ldots, a_{k-v+|\mathcal{S} \cap \mathcal{I}|}\} \cup \{b_1, \ldots, b_{j-u+|\mathcal{S} \cap \mathcal{I}^c|}\} \cup \{c_1, \ldots, c_k\} \cup \{d_1, \ldots, d_j\}) \tag{A.575}$$

$$p^{r/n} = \max_{\{\mathcal{I}: \mathcal{I} \subseteq [n], |\mathcal{I}| = n-r+1\}} p_{\mathcal{I}}, \tag{576a}$$

$$q^{r/n} = \max_{\{\mathcal{I}:\mathcal{I}\subseteq[n],|\mathcal{I}|=n-r+1\}} q_{\mathcal{I}}, \tag{577}$$

$$f_k = \max_{\mathcal{K}:|\mathcal{K}|=k} f(\{2p_i, i \in \mathcal{S}^- \cap \mathcal{K}^c\} \cup \{2q_i, i \in \mathcal{S}^+ \cap \mathcal{K}\}) \geq p^{k+1/n} > \alpha \tag{4.578}$$

$$f_k = \max_{\mathcal{K}:|\mathcal{K}|=k} f(\{2p_i, i \in \mathcal{S}^- \cap \mathcal{K}^c\} \cup \{2q_i, i \in \mathcal{S}^+ \cap \mathcal{K}\}) \geq q^{n-k+1/n} > \alpha \tag{A.579}$$

$$\begin{aligned} &\mathrm{Var}[a_{L+1}^j|x] \\ &= \mathrm{Var}[a_L^j + f_L^j(h_L)|x] \\ &= \mathrm{Var}[a_L^j|x] + \mathrm{Var}[f_L^j(h_L)|x] + 2\mathrm{Cov}[a_L^j, f_L^j(h_L)|x] \end{aligned} \tag{580a}$$

$$\begin{aligned} &\mathrm{Var}[f_L^j(h_L)|x] \\ &= \mathbb{E}[\mathrm{Var}[f_L^j(h_L)|h_L]] + \mathrm{Var}[\mathbb{E}[f_L^j(h_L)|h_L]]. \end{aligned} \tag{581}$$

$$\alpha\mathbb{E}[E_L(h_L)] \leq \mathbb{E}[\mathrm{Var}[f_L^j(h_L)|h_L]] \leq \beta\mathbb{E}[E_L(h_L)], \tag{1.582}$$

$$0 \leq \mathrm{Var}[\mathbb{E}[f_L^j(h_L)|h_L]] \leq MD \max_k \mathrm{Var}[a_L^k|x] \tag{A.583}$$

$$\left|\mathrm{Cov}[a_L^j, f_L^j(h_L)|x]\right| \leq \sqrt{M}D \max_k \mathrm{Var}[a_L^k|x] \tag{584a}$$

$$C_L\mathbb{E}[E(x_1, ..., h_{L-1})] \leq \mathrm{Var}[a_L^j|x] \tag{585}$$

$$\mathrm{Var}[a_L^j|x] \leq C'_L\,\mathbb{E}[E(x_1, ..., h_{L-1})]. \tag{5.586}$$

$$\mathrm{Var}[a_{L+1}^j|x] \geq C\,\mathbb{E}[E(x_1, ..., h_L)], \tag{A.587}$$

$$\begin{aligned} p(w) &= \mathcal{N}(w|0, \alpha^{-1}I), \\ p(\mathbf{y}|\mathbf{X}, w) &= \prod_{i=1}^N y_i^{\sigma(w^T x_i)}(1-y_i)^{1-\sigma(w^T x_i)}, \end{aligned} \tag{588a}$$

$$\begin{aligned} q(w) &= \mathcal{N}(w|\mu_{\mathrm{MAP}}, \Lambda^{-1}), \\ \Lambda &= \sum_{i=1}^N \sigma(w^T x_i)(1-\sigma(w^T x_i))x_i x_i^T + \alpha I, \end{aligned} \tag{589}$$